My
First
Bitcoin

Bitcoin Diploma

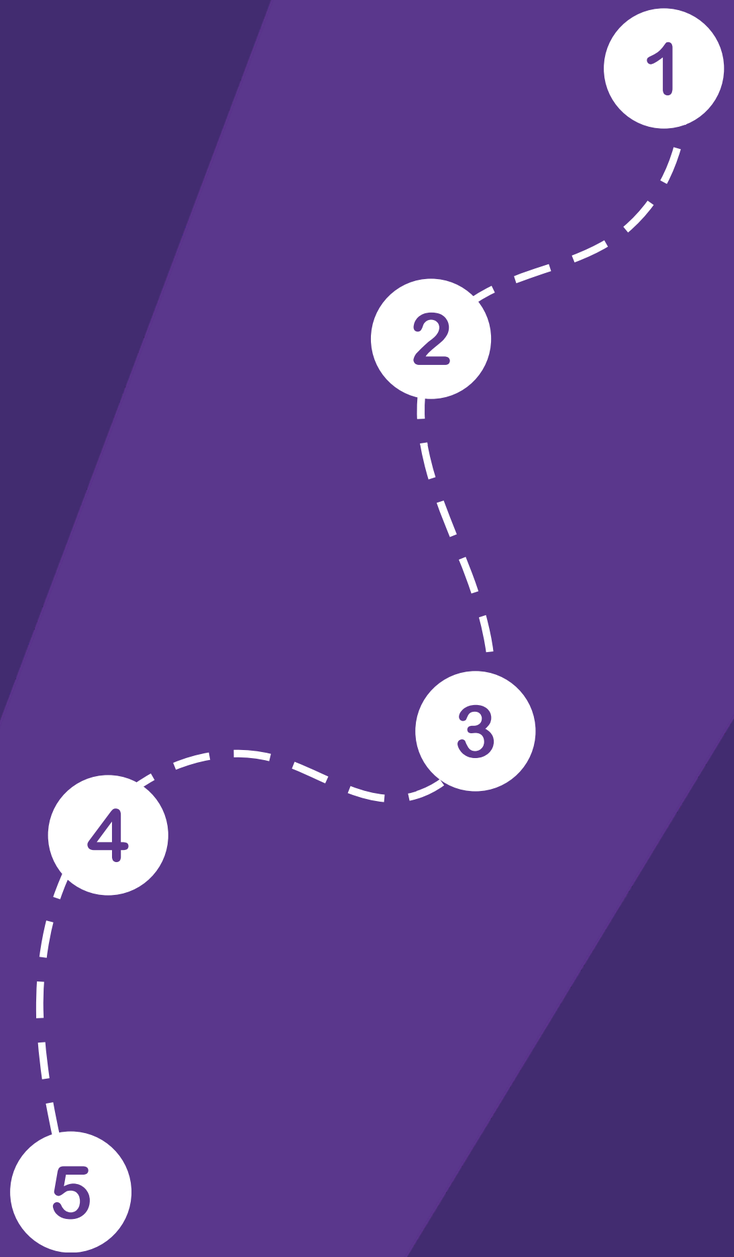
Meet your instructor

Meet your instructor

Gerardo Moran

Salvadoran

Studying and teaching Bitcoin since 2023



Roadmap

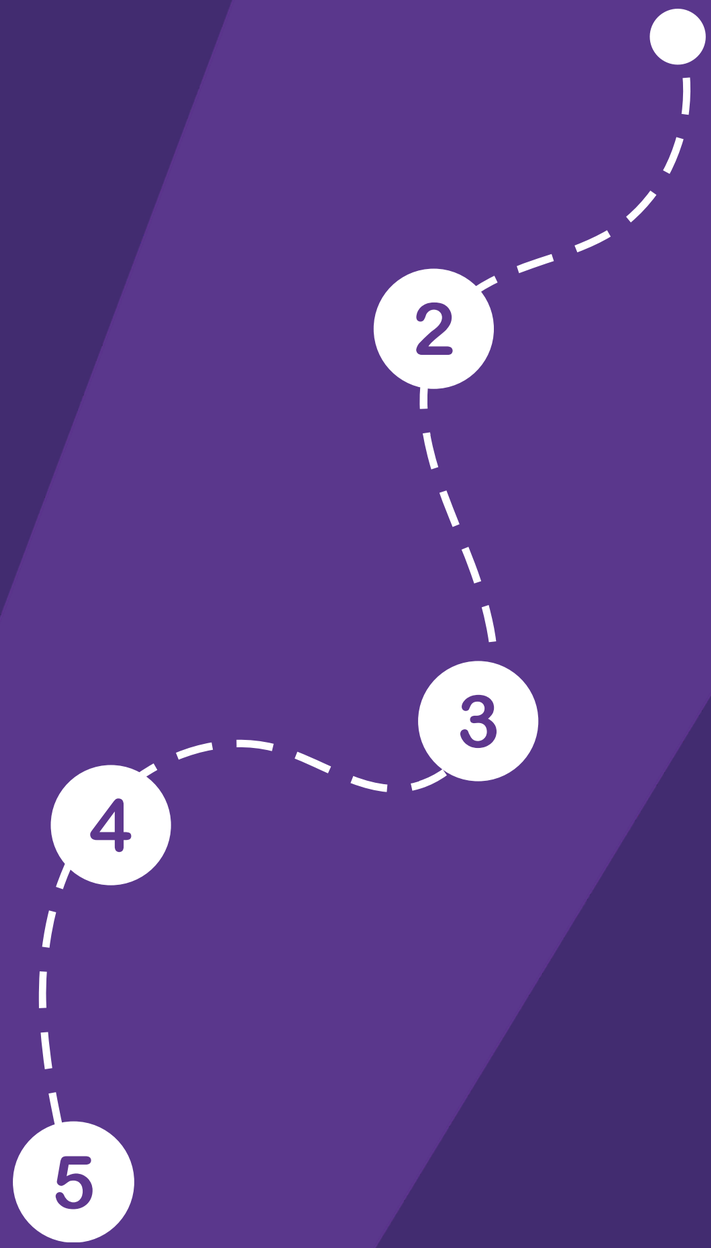
Week 1

Session 1

What Is Money?

Session 2

The History of Money



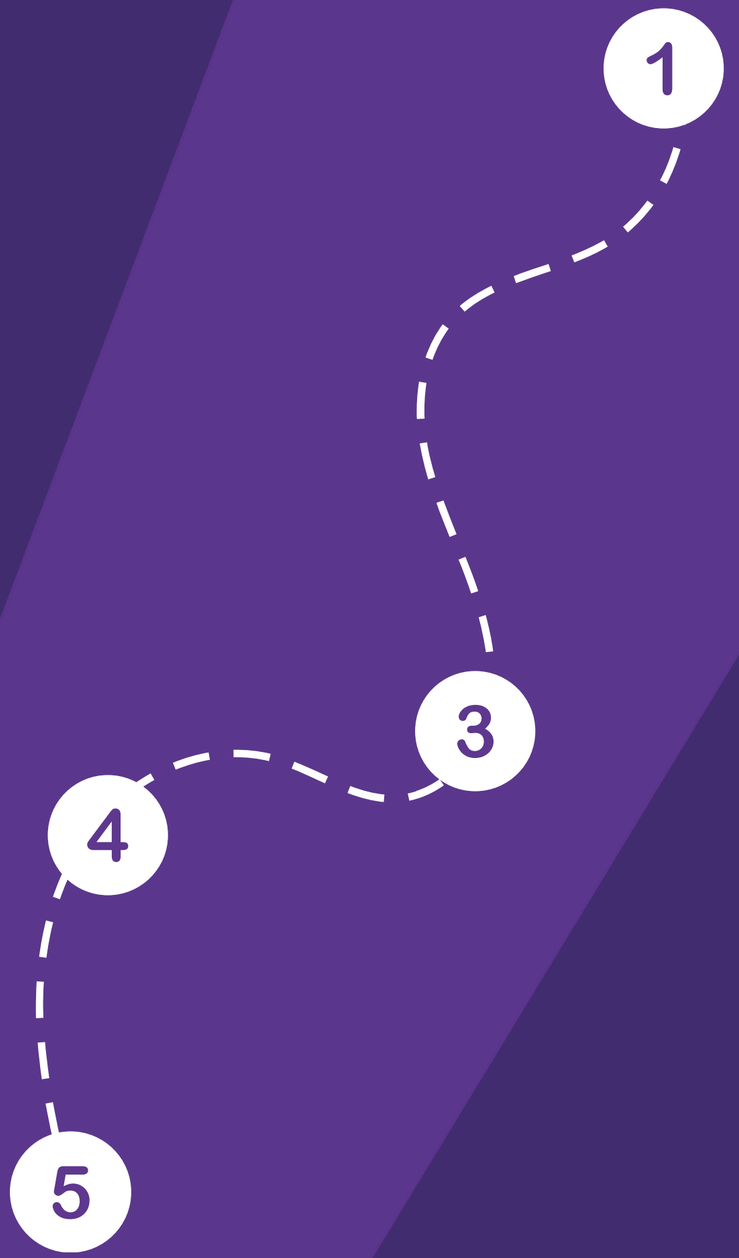
Week 2

Session 3

What Is Fiat Money?

Session 4

How Problems Lead to Solutions



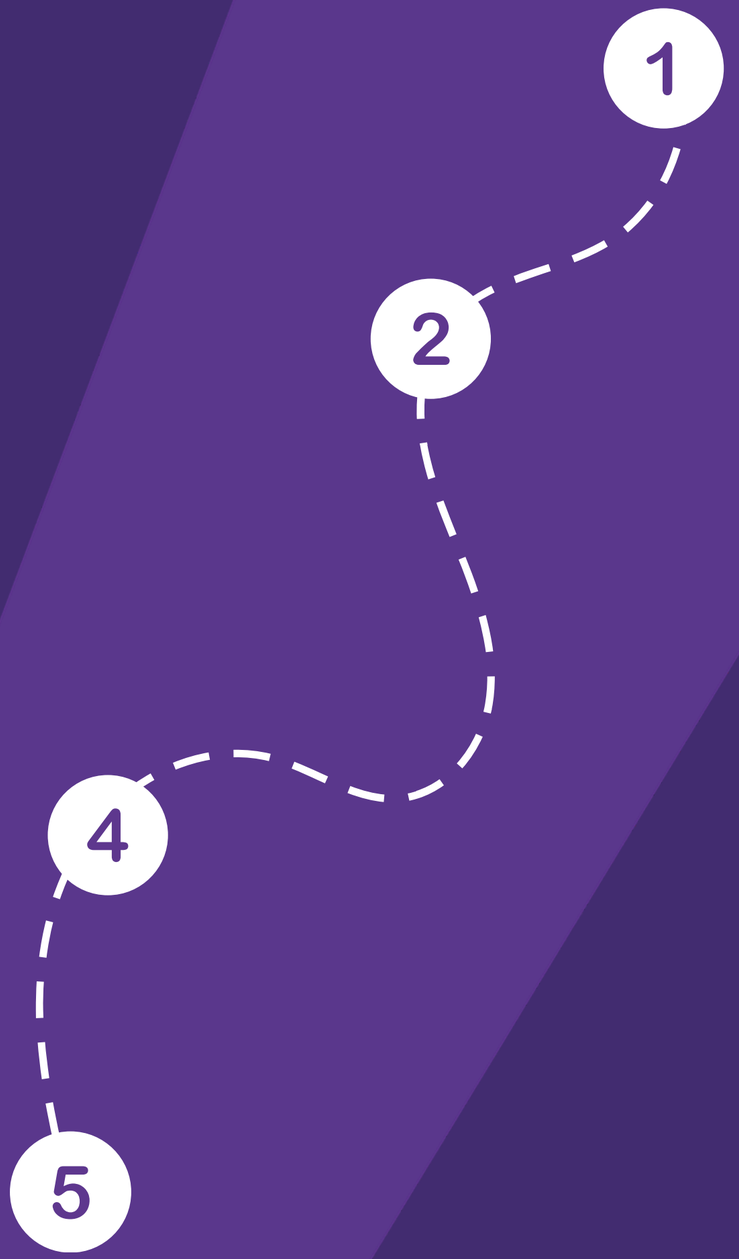
Week 3

Session 5

What Is Bitcoin?

Session 6

How to Use Bitcoin



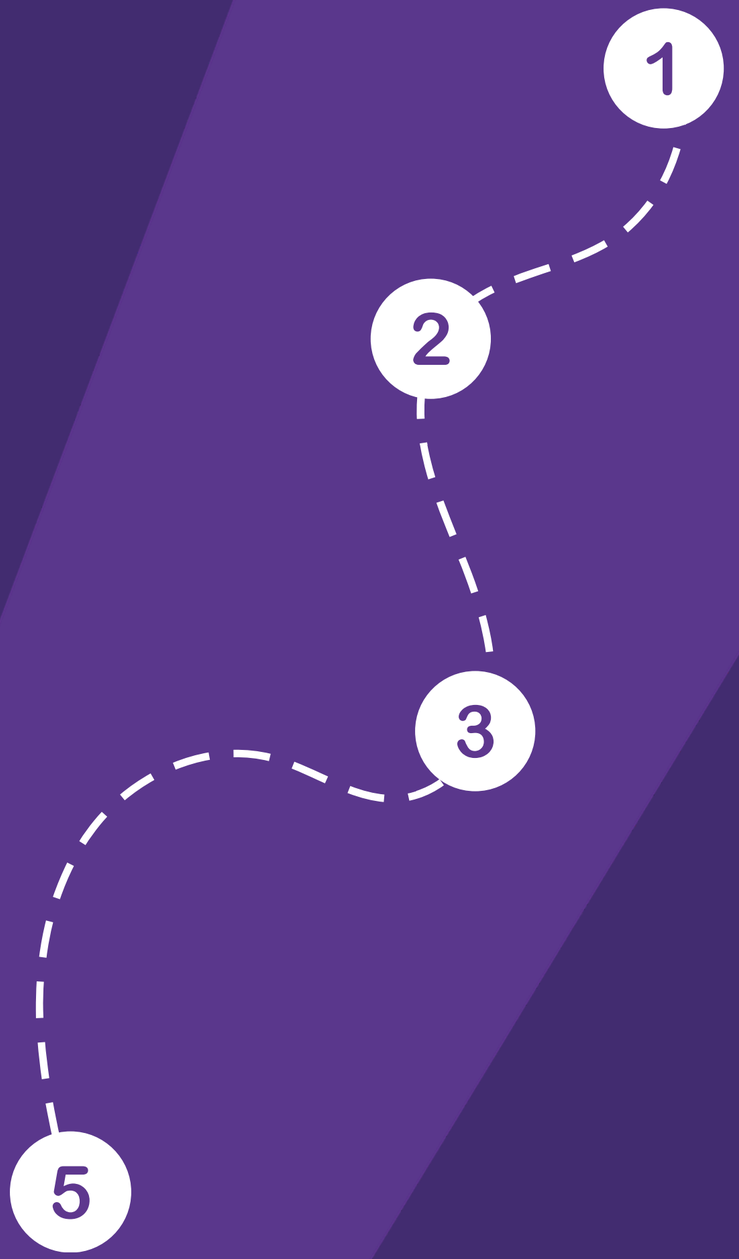
Week 4

Session 7

Using Bitcoin in Daily Life

Session 8

How Bitcoin Works



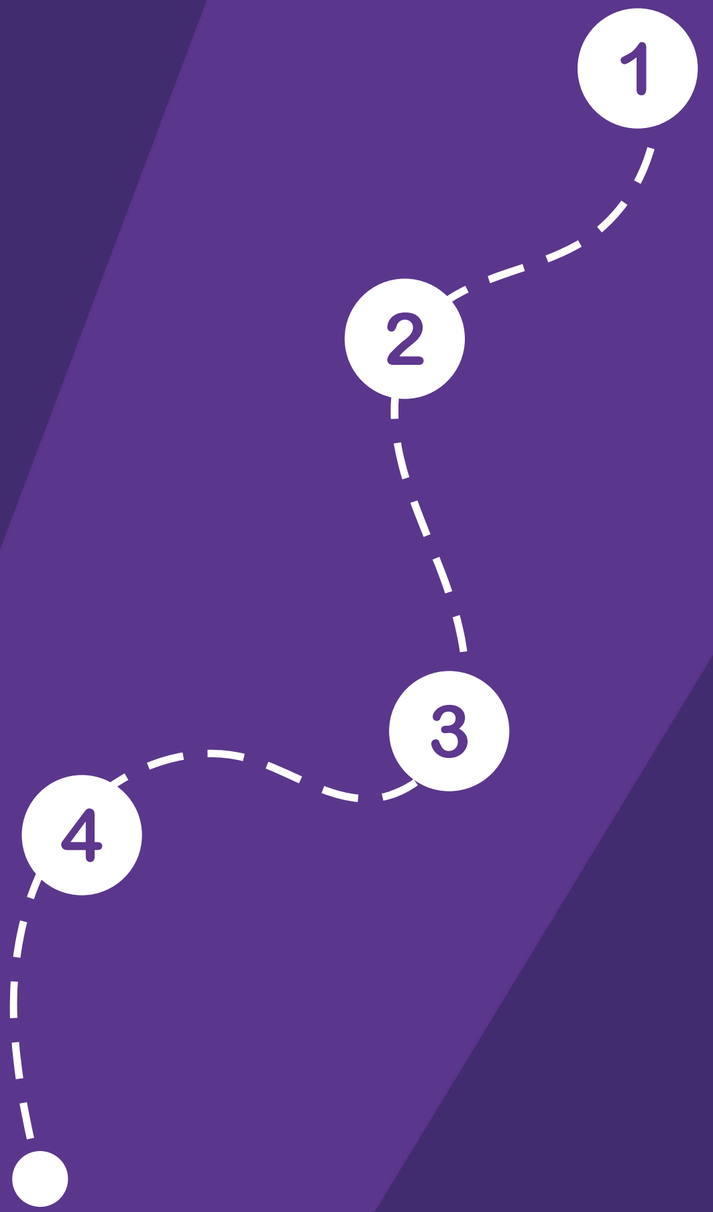
Week 5

Session 9

How Does Bitcoin Mining Work

Session 10

What Future can Bitcoin Build



Certificate of participation

What Is Money?

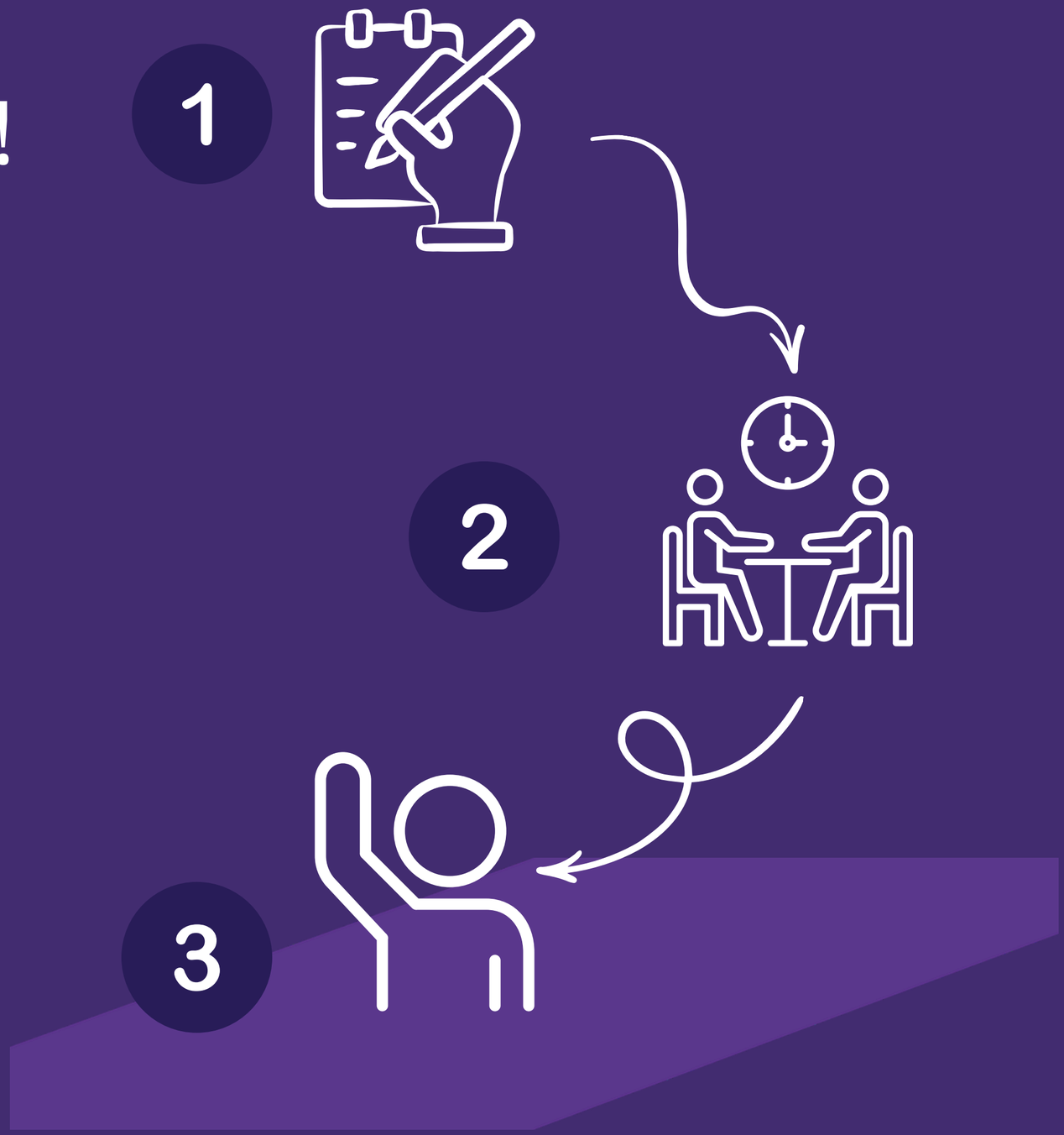
Module 01

Let's get into an activity!

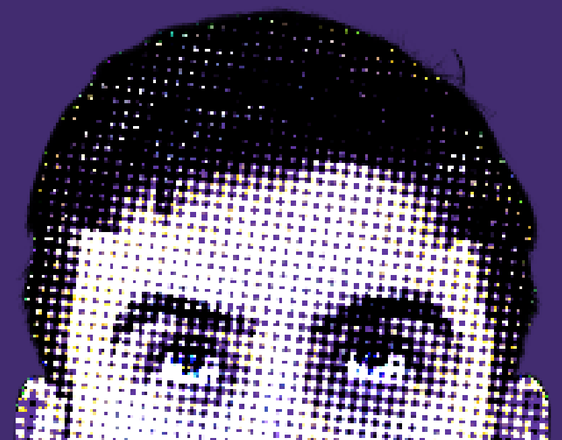
1. Take note of the 4 questions.

2. Think about your own answers to these questions within the 5-minute time limit.

3. Share your answers.



4 Questions about money



- Why do we need money?
- What is money?
- Who controls money?
- What gives money its value?

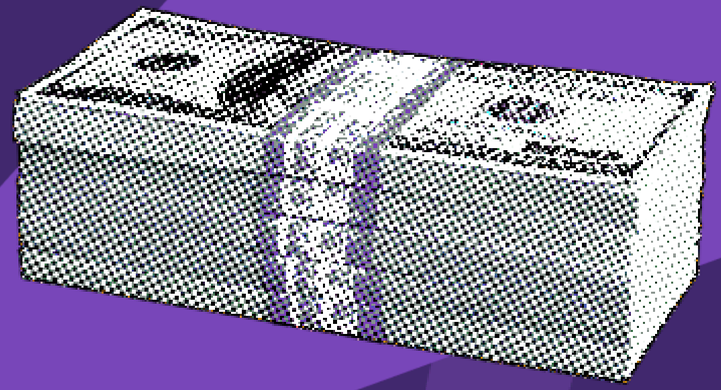
“Money is a guarantee that we may have what we want in the future. Though we need nothing at the moment, it ensures the possibility of satisfying a new desire when it arises.”

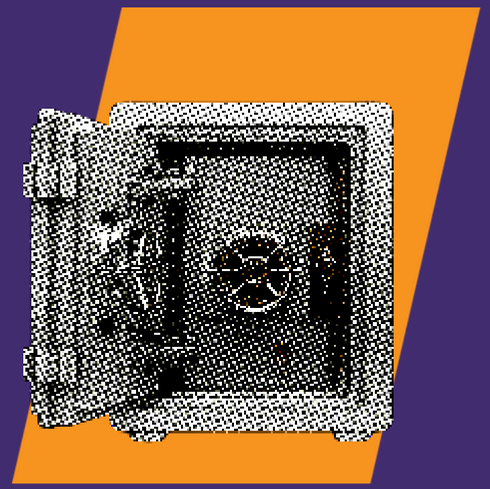
— Aristotle

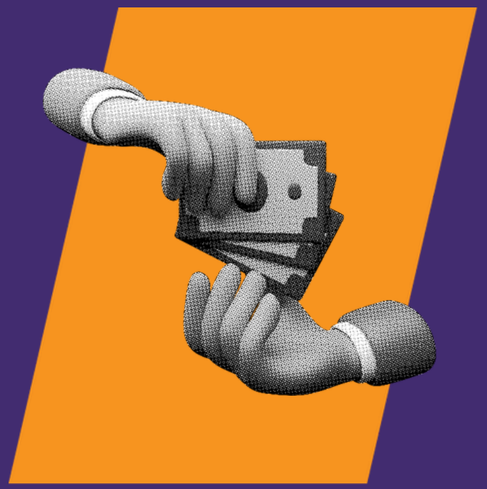
Definition of money

Definition of money

Money facilitates trade because everyone accepts it as final payment. It also allows us to measure value and compare different goods and services.

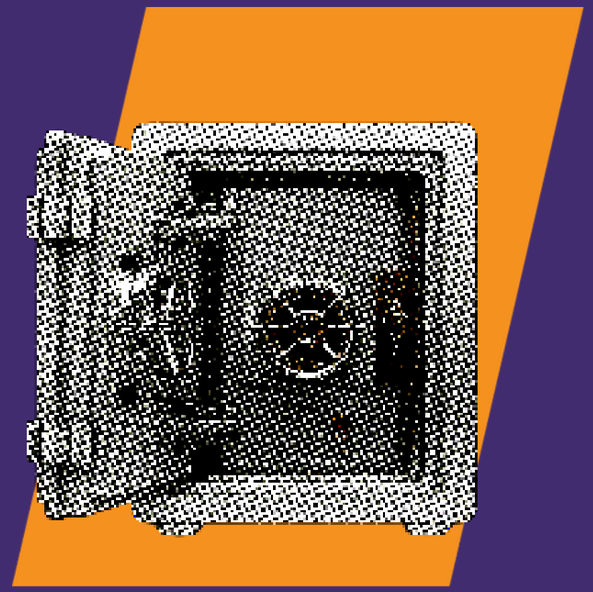






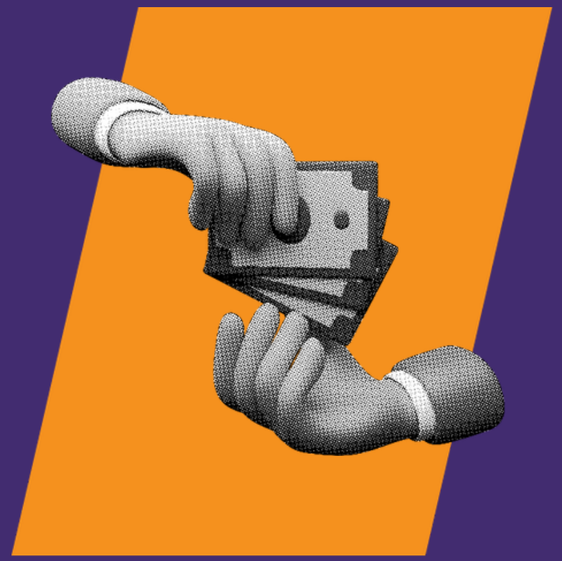
Functions of Money

Store of Value

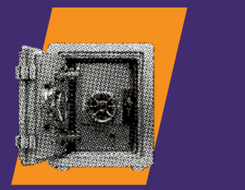


Money **should maintain** its value over time, making it useful as a method to save and invest the value of human labor. This lets people use money to plan for the future.

Medium of Exchange

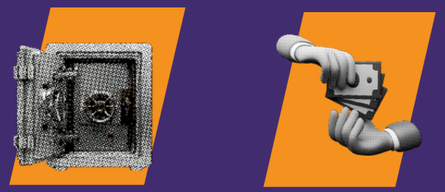


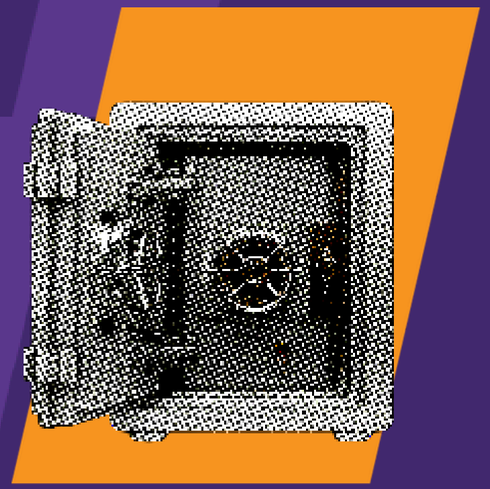
With money, you don’t have to find someone who wants exactly what you have to trade. Instead, you can use money to buy and sell anything you want. This makes trading and commerce much more **convenient and efficient.**



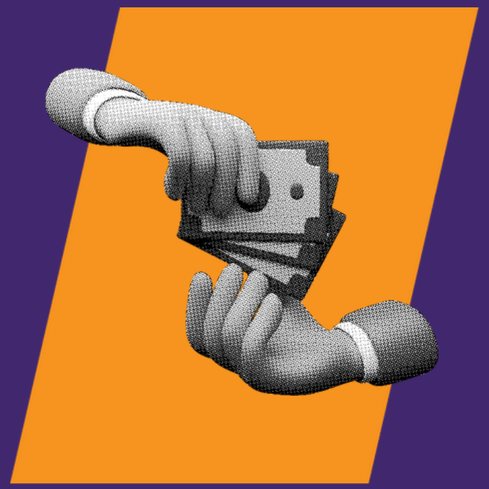
Unit of Account

Money provides a **universal standard for measuring** value that allows people to express and compare the price of different goods and services. This allows for a more efficient and transparent market, where people can make informed decisions about what to buy and sell.





Store of Value



Medium of Exchange

Unit of Account

Properties of Money

Durable

Refers to money's ability to **resist physical deterioration** and last over time. This ensures that money can circulate in the economy in an acceptable and recognizable state.

Divisible

Refers to money's ability to be **divided into smaller units** so people can use it to make purchases of varying amounts.

Portable

Refers to the ease with which money can be **transported and carried around.**

Refers to the interchangeability of money so that one unit of money is **equivalent to another unit** of the same value.

Uniform

Refers to the widespread acceptance of money as a form of payment so that people can use it to buy and sell goods and services with confidence.

Acceptable

Refers to how difficult it is to make more units of the money, which helps maintain its value and prevents us from having to spend more money to buy the same amount of goods.

Scarcity

Properties of Money

Let's get into an activity!

Use the following questions to help determine how well the different items on screen meet the properties of money.

Diamonds

Local currency

Cows

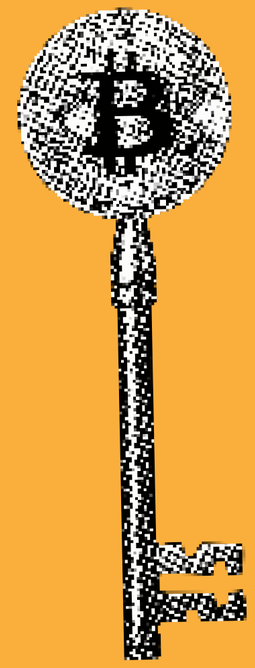
Hot Sauce

- **Durability:** Can the money withstand wear and tear over time?
- **Portability:** Can the money be easily transported and used in different locations?
- **Fungibility:** Is the money interchangeable with other forms of money?
- **Acceptability:** Is the money widely accepted as a form of payment?
- **Scarcity:** Is the money scarce and difficult to make more of?
- **Divisibility**: Can the money be divided into smaller units?

Time to share your answers

Key takeaways

1. Definition of Money
2. Functions of Money
3. Properties of Money



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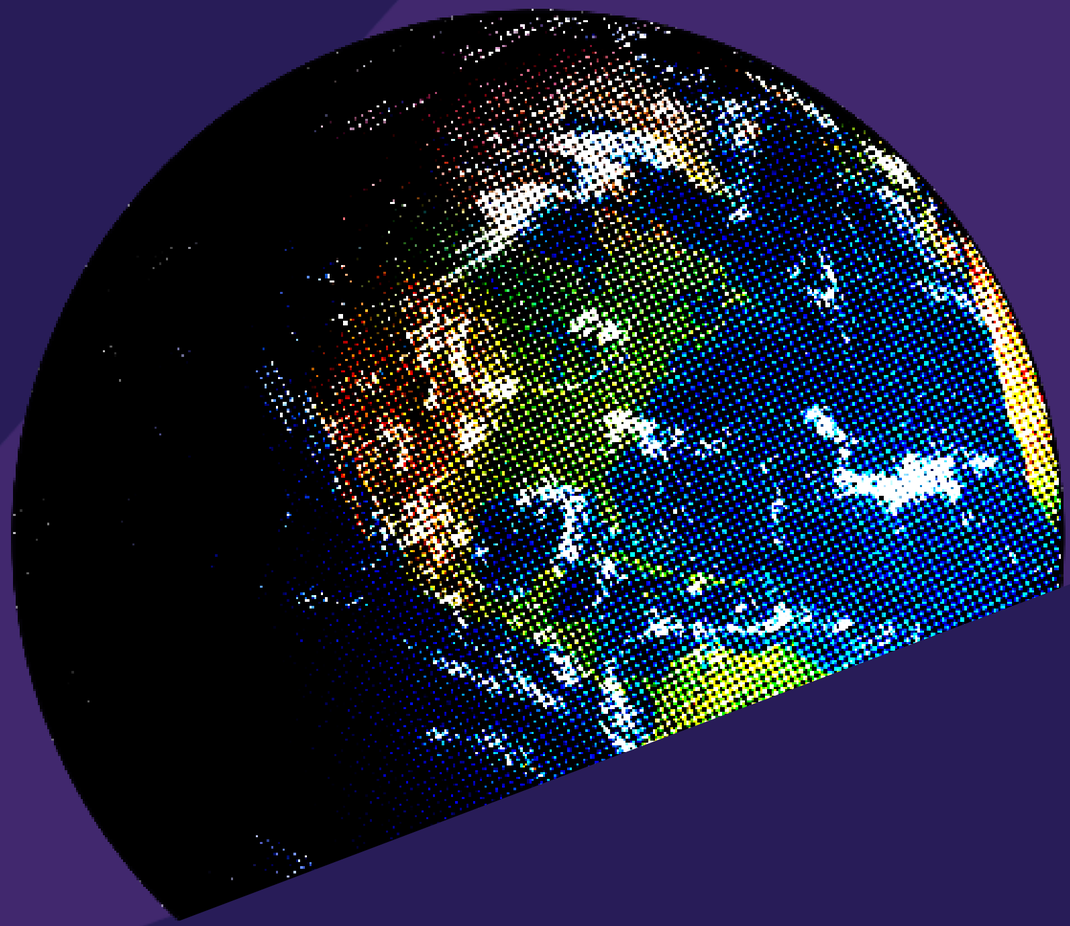
The History of Money

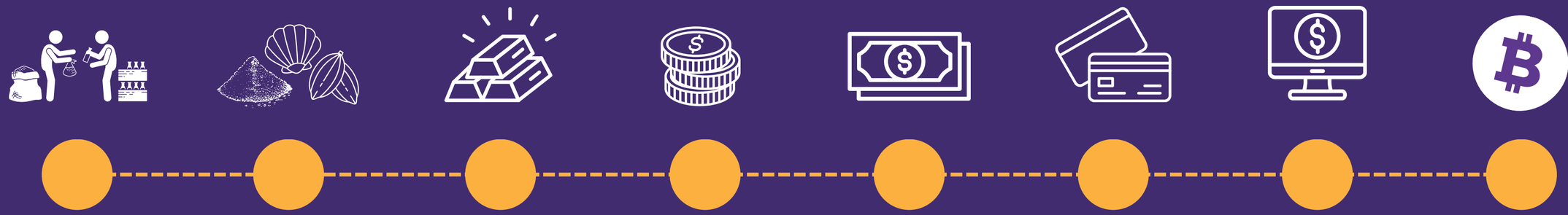
Module 02

What was the first form of money in your country?



“Money was not evolved by design but arose out of the market process. It was not created by governments. It emerged over time as a spontaneous order.”

— Murray Rothbard



From Barter to Modern Currency

Barter

In a barter economy, people trade goods and services directly with each other. For a trade to happen, each person must have something the other wants.

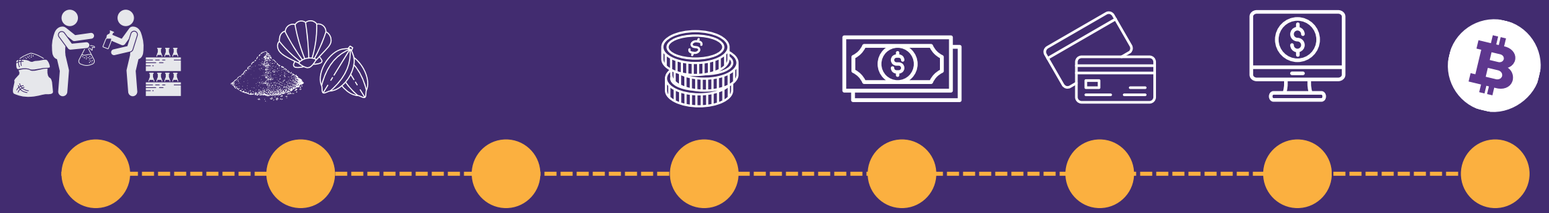
Commodity money

One of the first forms of money as a medium of exchange and with its own intrinsic value.

Development of Coinage and Paper Money

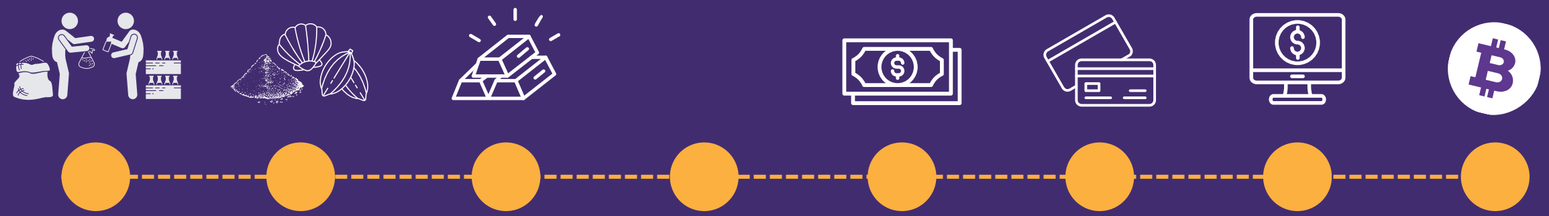
Precious Metals

Gold and silver used as a medium of exchange due to their durability, scarcity, and general acceptance.



Coins

Coins began to be used as an alternative to bartering and other forms of exchange.



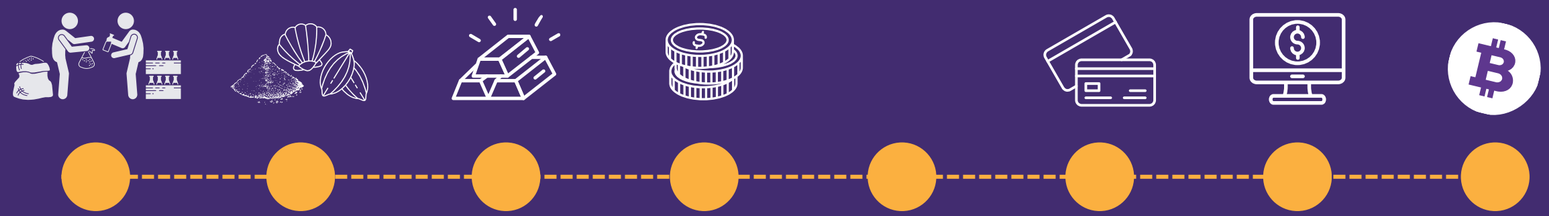
Representative money

Paper certificates backed by gold and other precious metals, practical to carry and exchange.

Transition from Sound to Unsound Money

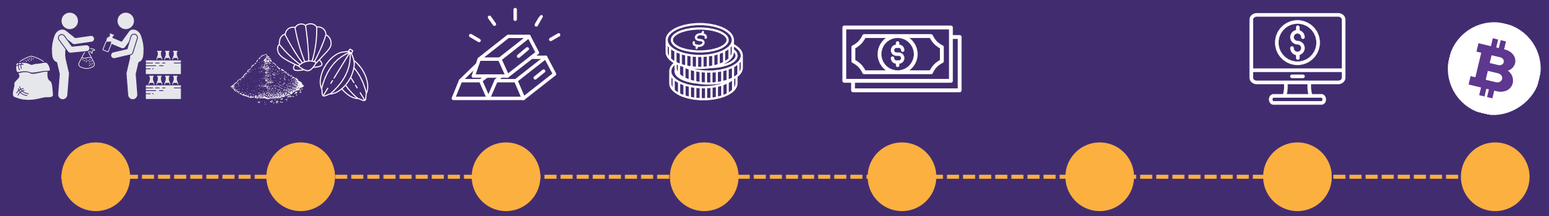
Fiat money

The fiat system is characterized by its mandatory nature, being imposed on people through legal tender laws. The Latin term fiat means “by decree” and thus refers to a directive issued by some authority.



Paper to Plastic

We’ve come a long way from the introduction of the first credit card back in the 1950s. Today, with a simple swipe or a contactless tap, we can make our purchases whenever we want, without any hassle.



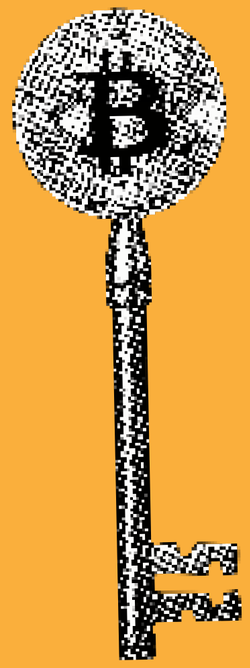
Digital Currency

Unlike traditional monies, digital currencies exist solely in electronic form. They are stored and exchanged using computers and special software. Much like how email allows us to send and receive messages instantly and without shipping costs, digital currencies make it possible to send and receive value instantaneously and at very little cost.

In today's session, we took a journey through the history of money and discovered how humanity has always sought alternatives to solve the problems of each previous system. Today, fiat money faces its own challenges and limitations, and history shows us that it is no exception to the rule. This leaves us with a key question for the future: Does a definitive alternative already exist?

Key takeaways

1. History of money
2. Does a definitive alternative already exist?



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What is Fiat Money?

Module 03

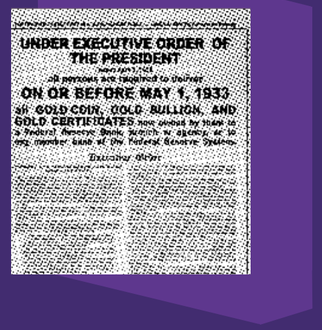
“The history of mankind is the history of money losing value.”

— Milton Friedman

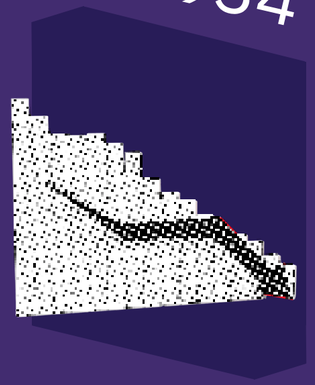
Brief History of Fiat Money

1913

1933

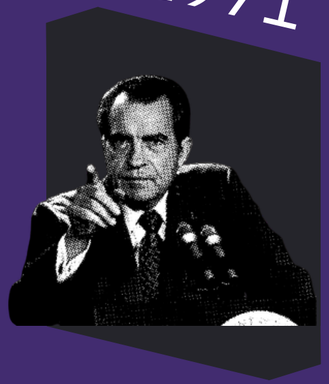


1934



1944

1971



1980

Brief History of Fiat Money

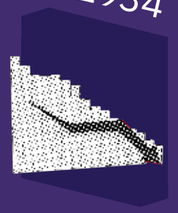
Creation of “The Federal Reserve”

1913

In 1913, the United States created the Federal Reserve, a central bank that could issue new money and support struggling banks.

1923

1934



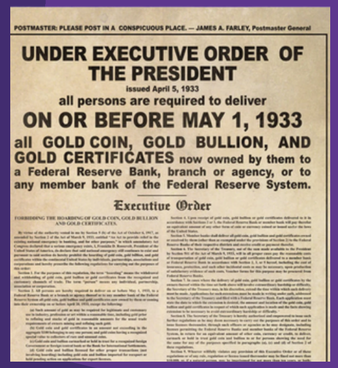
1944

1971

1980

Executive Order 6102

1933



POSTMASTER: PLEASE POST IN A CONSPICUOUS PLACE. — JAMES A. FARLEY, Postmaster General

UNDER EXECUTIVE ORDER OF THE PRESIDENT

issued April 5, 1933

all persons are required to deliver

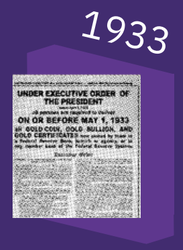
ON OR BEFORE MAY 1, 1933

all GOLD COIN, GOLD BULLION, AND GOLD CERTIFICATES now owned by them to a Federal Reserve Bank, branch or agency, or to any member bank of the Federal Reserve System.

Executive Order

FORBIDDING THE HOARDING OF GOLD COIN, GOLD BULLION AND GOLD CERTIFICATES.

During the 1930s, the U.S. government also required citizens to surrender their gold in exchange for paper dollars.



Gold Reserve Act

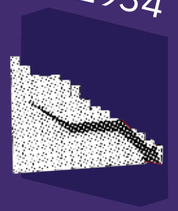
In just 1 year, the value of the dollar was devalued

1 oz = $20.67

1923

1933

1934



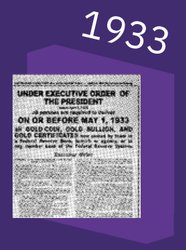
1971

1980

Bretton Woods Agreement

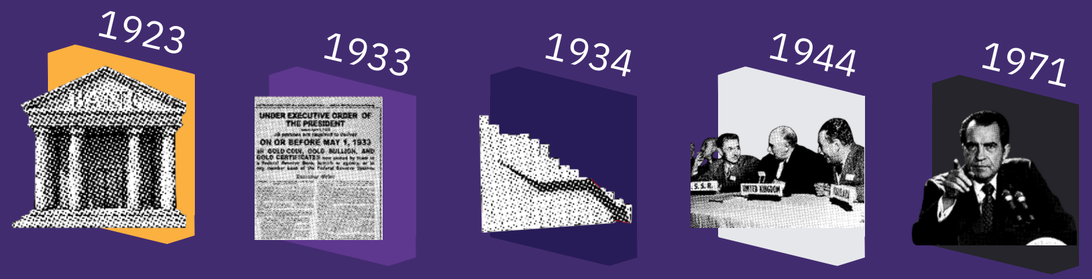
1944

After World War II, the Bretton Woods system linked global currencies to the U.S. dollar, and the dollar could still be exchanged for gold.



Nixon Shock

This system ended in 1971, when the United States stopped allowing dollars to be redeemed for gold. Since then, most countries use fiat money.



Money lost 96% of its value

1980

Fiat money is currency that is not backed by a physical commodity like gold. Instead, it has value because governments declare it legal tender and people trust and accept it for payments. Since then, most countries use fiat money.

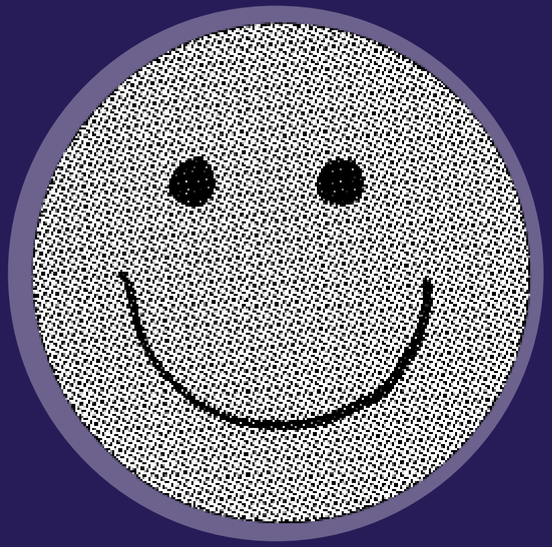
“The root problem with conventional currency is all the trust that’s required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust.”

— Satoshi Nakamoto

Pros of Fiat Money

Ease of use: Fiat money is convenient for everyday transactions.

Lower costs and risks: Fiat money doesn’t require heavy security like gold, making it cheaper and safer



Inflation risks: Governments can print fiat money at will, devaluing the currency and causing prices to rise, which decreases the purchasing power of savers

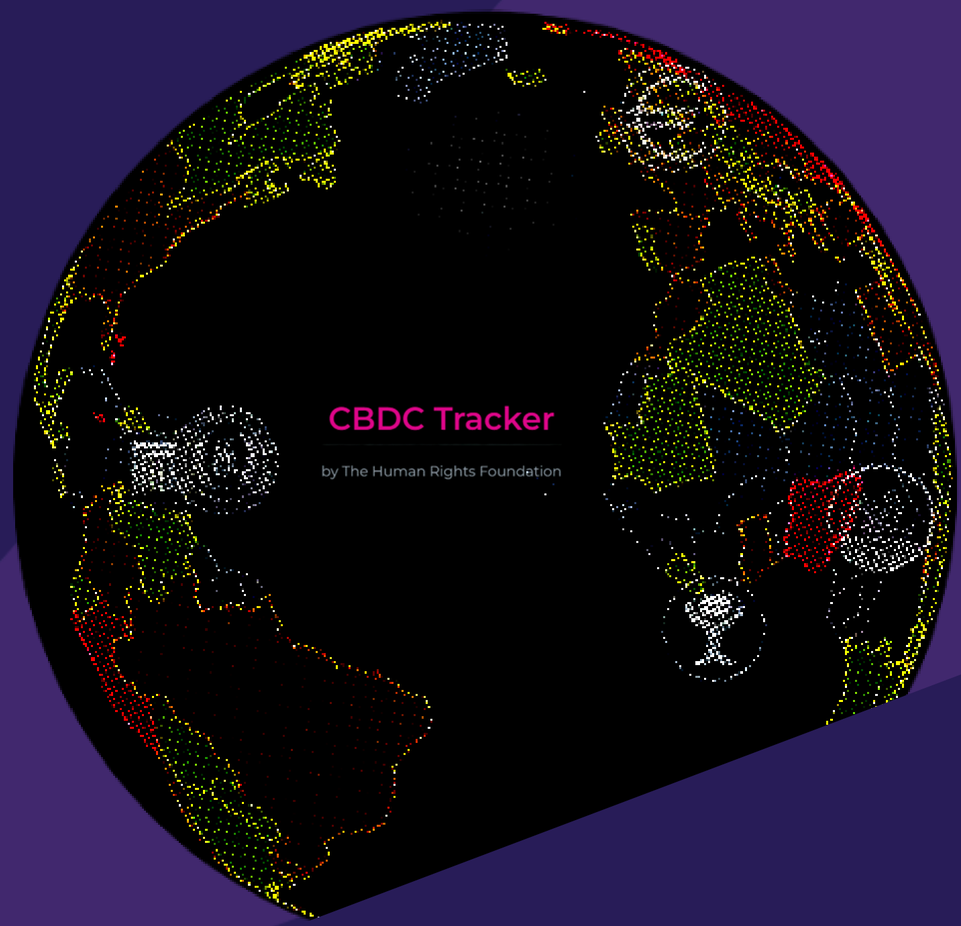
Centralized control and manipulation: Small groups can influence and manipulate the system, leading to politically motivated debanking and confiscation.

Counterparty risk: If the government faces challenges, and the public loses trust, the currency can lose value

Cons of Fiat Money

Central Bank Digital Currencies

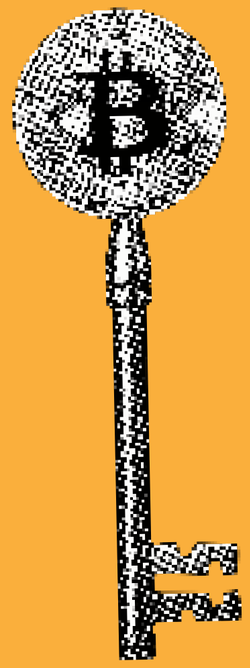
While CBDCs seem to be the future of fiat money, the world's current monetary system already oper- ates on a pure fiat standard. Fiat currencies are no longer tied to gold, resulting in a significant expansion of the monetary supply without any real restriction.



Key takeaways

1. Fiat money is currency not backed by a physical commodity.

2. The fiat system relies heavily on trust and debt, often leading to cycles of credit expansion, booms, and financial crises.



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How Problems Lead to Solution

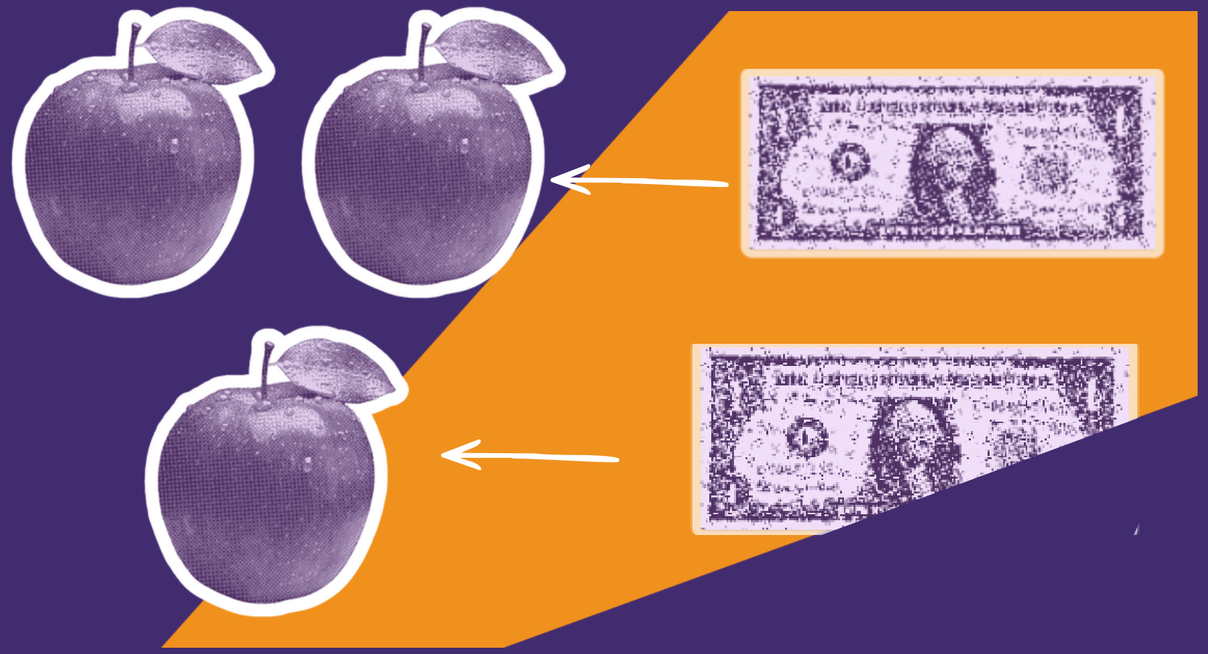
Module 04

Whoever controls the volume of money in our country is absolute master of all industry and commerce... when you realize that the entire system is very easily controlled, one way or an- other, by a few powerful men at the top, you will not have to be told how periods of inflation and depression originate.

—James A. Garfield, US President

Money Buys Less

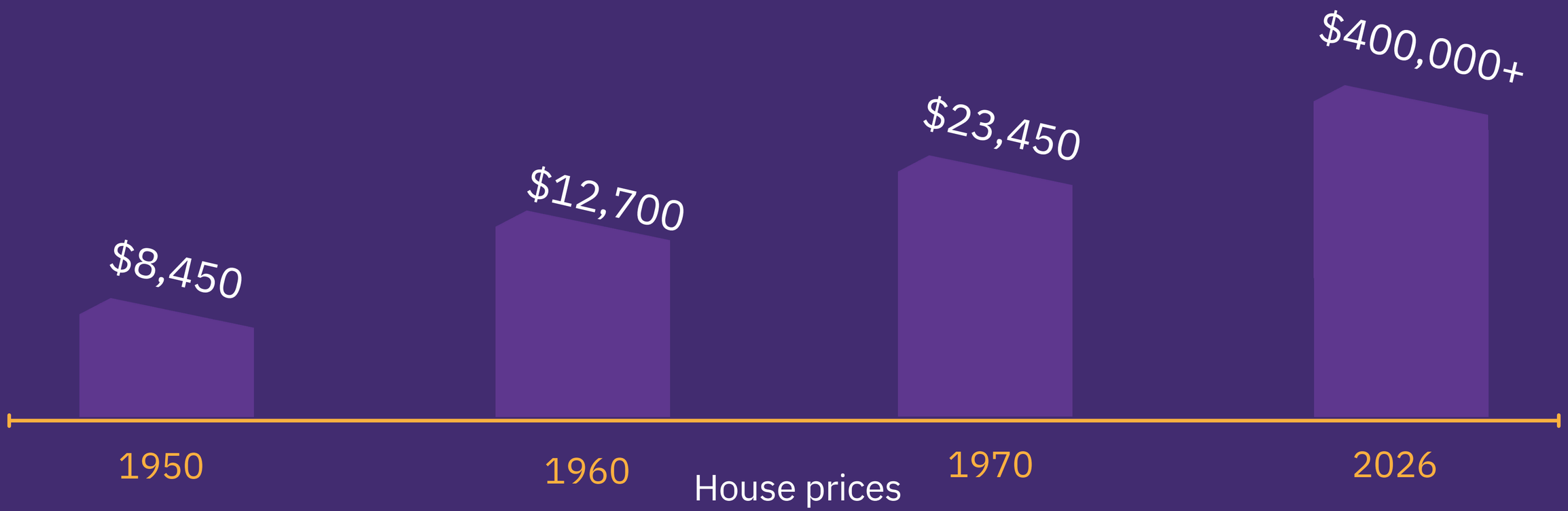
Loss of purchasing power



As money loses its value over time, we need more and more bills to reach the true value of a product.

How many houses can you buy with $20,000 dollars nowadays?

How many houses can you buy with $20,000 dollars nowadays?



Consequences of the Fiat System

Consumerism and immediate debt

Inflation destroys long-term savings, forcing people to live in debt and focus only on surviving day to day.

Government power and wars

Unlimited money printing centralizes political power and allows superpowers to fund endless conflicts and wars.

Wealth concentration and inequality

Newly printed money benefits the elites first before inflation hits, enriching a few while impoverishing the majority.

Consequences of the Fiat System

“The computer can be used as a tool to liberate and protect people, rather than to control them.”

— Hal Finney

The Cypherpunks

The Cypherpunks

They anticipated both the potential and the danger of the personal computer, whether as a tool to enable freedom to empower the individual, or as a tool for complete control and surveillance.

The term "Cypherpunk" is a fusion of "cypher," referring to cryptographic code, and "punk," representing the countercultural ethos of rebellion

The Cypherpunks

Group of activists

Cryptographers and Programmers

Privacy advocates

United by a common vision:

The search for privacy, security, and a decentralized digital future.

Centralized

vs

Descentralized

Centralized

- **Single point of failure:** If the central authority fails or suffers a cyberattack, the entire system can collapse.
- **Control and censorship:** A small group holds all the power, limits user autonomy, and can block access to resources.
- **Inefficiency and hidden metrics:** Unnecessary middlemen make operations slow and expensive, while the lack of transparency destroys trust.

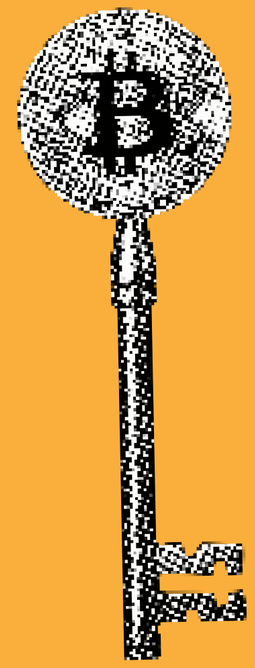
Descentralized

- **High resilience and security:** With no single point of failure, the system remains reliable, strong, and highly resistant to attacks.
- **Total sovereignty and privacy:** Users have complete control over their money and data, protected by distributed information.
- **Transparency and equal access:** Everyone sees the same information and anyone can join without permissions, creating a fairer system.

Key takeaways

1. Increasing the money supply reduces purchasing power and causes prices to rise over time.

2. Technologists and Cypherpunks began working on decentralized digital money to solve these problems.



My
First
Bitcoin

What is Bitcoin?

Module 05

Did Bitcoin just appear overnight?

“A lot of people automatically dismiss e-currency as a lost cause because of all the compan- ies that failed since the 1990s. I hope it's obvious it was only the centrally controlled nature of those systems that doomed them. I think this is the first time we’re trying a decentralized non-trust-based system.”

— Satoshi Nakamoto

The Creation of Bitcoin

E- cash (1982)

DigiCash (1990)

B-Money (1996)

Hash Cash (1998)

Bit Gold (1998)

e-Gold (2004)

E- cash (1982)

The original cryptographic money concept created by David Chaum. It failed because **it required traditional banks to adopt its centralized software**, and they showed little commercial interest.

DigiCash (1990)

The commercial version of E-cash implemented as a company. It failed because **it relied on a 100% centralized infrastructure**; the company went bankrupt in 1998 due to poor market management.

B-Money (1996)

A theoretical proposal by Wei Dai that introduced money creation through solving logical puzzles. It failed because **it never left the paper stage**, as it could not define an efficient consensus mechanism without intermediaries.

Hash Ca (1998

A mechanism crea Adam Back to limi spam using "Pro Work." It failed bec **was not designe currency but as a spam tool**; it lac balance-tracking s

B-Money (1996)

A theoretical proposal by Wei Dai that introduced money creation through solving logical puzzles. It failed because **it never left the paper stage**, as it could not define an efficient consensus mechanism without intermediaries.

Hash Cash (1998)

A mechanism created by Adam Back to limit email spam using "Proof of Work." It failed because it **was not designed as a currency but as an anti-spam tool**; it lacked a balance-tracking system.

Bit Gold (1998)

Nick Szabo's proposal to create scarce digital property similar to physical gold. It failed because **it remained theoretical**; it could not solve the "double-spending" problem (preventing someone from spending the same coin twice) in a decentralized way.

e-Gold (2004)

A massive, private digital currency system backed by real gold stored in vaults. It failed because **it had a physical and legal single point of failure**; the U.S. government intervened, froze its accounts, and shut it down.

The Creation of Bitcoin

E- cash (1982)

DigiCash (1990)

B-Money (1996)

Hash Cash (1998)

Bit Gold (1998)

e-Gold (2004)

The Creation of Bitcoin

Cryptography

Proof of Work

Digital Scarcity

Double-Spending Prevention

Distributed Ledger

Peer-to-Peer Consensus

Blinded Signatures

Unforgeable Costliness

Bitcoin

2008

A Peer-to-Peer Electronic Cash System

On October 31st, 2008, Satoshi Nakamoto published the Bitcoin Whitepaper titled: 'Bitcoin: A Peer-to-Peer Electronic Cash System'.

- Context: This was the moment the theory and the solution to the double-spending problem were presented to the world.

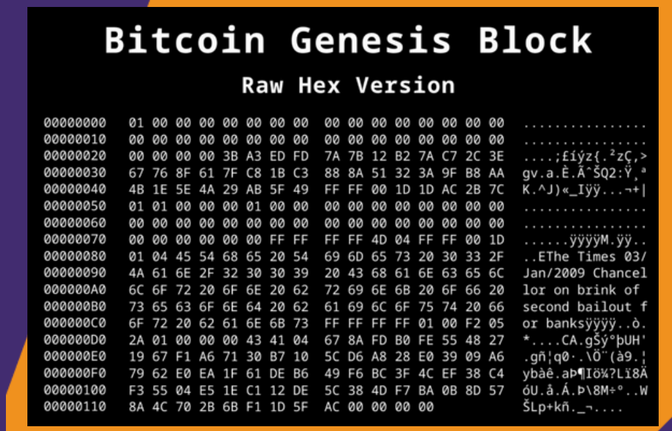
Satoshi Nakamoto

A visionary mind by the name of “Satoshi Nakamoto.” This anonymous person (man, woman, or group), long before Bitcoin, was engaging in discussions on cryptography and computer science on hacker forums to find practical solutions to replace the fiat system.

The Genesis Block

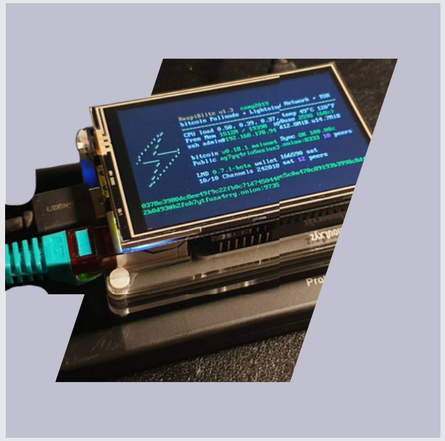
The Bitcoin network was officially launched on January 3rd, 2009, when Satoshi mined the 'Genesis Block' (Block 0).

Shortly after mining the first blocks and ensuring the system was stable, Satoshi Nakamoto gradually stepped back and eventually disappeared completely, leaving the project in the hands of the community.



The Players of the Game

Miners



Nodes

Users

Developers

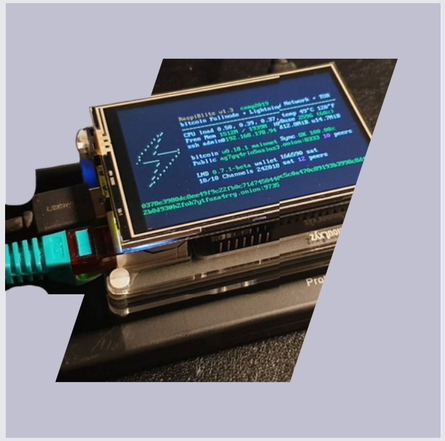
Miners

Miners are the backbone of Bitcoin. They work behind the scenes to maintain and secure the network through a mechanism called Proof-of-Work (PoW). These players are armed with special computers that boast heavy computational power.

They make this power available to the Bitcoin network, competing with each other in a worldwide lottery to add new blocks of transactions to Bitcoin’s decentralized ledger (the blockchain).

The Players of the Game

Miners



Nodes

Users

Developers

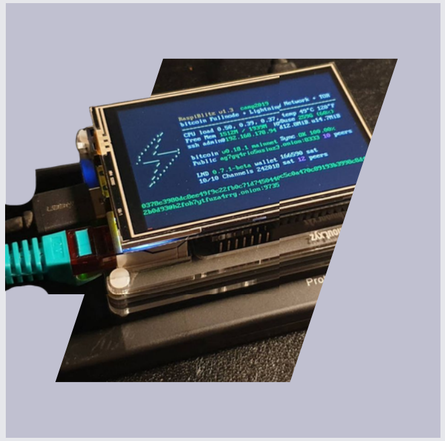
Nodes

Bitcoin nodes are run by ordinary people across the planet. These participants serve as the Bitcoin network's gatekeepers by running Bitcoin software on their computers on which they keep a copy of the entire ledger.

Nodes validate transactions and ensure that all participants adhere to the consensus rules.

The Players of the Game

Miners



Nodes

Users

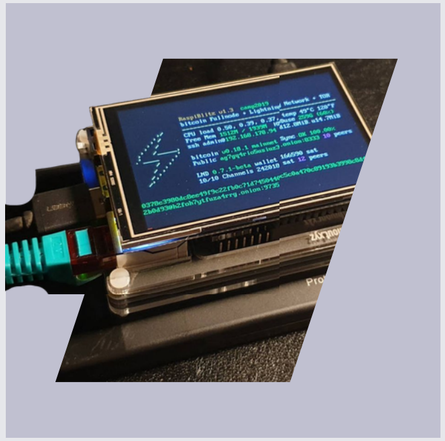
Developers

Users

Users — the lifeblood of the Bitcoin network — are individuals who engage in transactions. You can think of users as regular people who just have empowered themselves by integrating Bitcoin into their lives. For example, some users save their money in bitcoin while others use it as money to buy groceries and receive their salary.

The Players of the Game

Miners



Nodes

Users

Developers

Devs

Developers wield their technical expertise to enhance and innovate on the Bitcoin protocol.

These individuals contribute code, propose improvements, and address vulnerabilities, ensuring the network evolves in response to all types of challenges.

Bitcoin as Sound Digital Money

Principles of Bitcoin

Decentralized

No one controls it. A global network keeps it running

Peer-to-Peer

People send money directly to each other without banks.

Finite

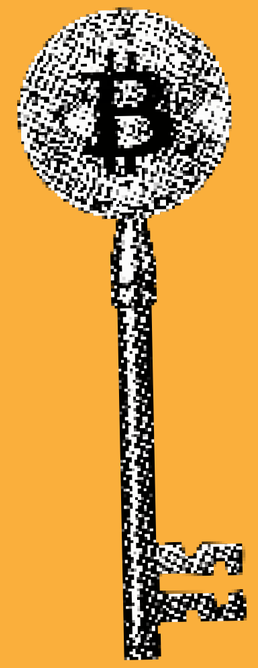
There will only ever be 21 million bitcoins.

In the simplest terms, Bitcoin is money. Bitcoin is not an investment but rather a safe, empowering way of saving your hard-earned money. Holding bitcoin won’t make you rich because it won’t give you a return of more bitcoin. Its value, measured against any fiat currency, does go up; but this is only because of its growing adoption and the devaluation of fiat currencies.

Key takeaways

1. The Bitcoin network runs on shared rules called consensus, ensuring no one can change the system or create more than 21 million bitcoin.

2. Different participants secure and maintain the network, including miners, nodes, users, and developers.



My
First
Bitcoin

How to Use Bitcoin

Module 06

“Why would anyone trust nerd money vs. central bank money? Nerds brought you the internet. Banks brought you the Great Depression. ”

— Satoshi Nakamoto

Acquiring Bitcoin

Acquiring Bitcoin

Get paid in bitcoin in exchange for your work and pay for other people’s products and services with bitcoin

Get paid in bitcoin

Acquiring Bitcoin

Mine bitcoin

Through mining, bitcoins are obtained by processing transactions and securing the network, receiving the block reward and user transaction fees in return.

Acquiring Bitcoin

Peer-to-Peer

P2P transactions allow the transfer of bitcoin directly between two users without banking intermediaries, either in person by scanning QR codes face-to-face, or online by sending funds over the internet using a digital address or invoice.

Centralized Exchanges

Centralized exchanges make buying bitcoin easy, but they require your identity, can misappropriate funds, and control your money.

The real risk is storing your coins there, so you must always transfer them to your own wallet. Additionally, there is no central bank in Bitcoin to print money and bail them out if they collapse.

Introduction to Bitcoin Wallets

Introduction to Bitcoin Wallets

Unlike physical money, bitcoin are not actually contained in a Bitcoin wallet. Instead, they live on the distributed ledger that the Bitcoin network constantly verifies and secures. So, how can you own bit- coin?

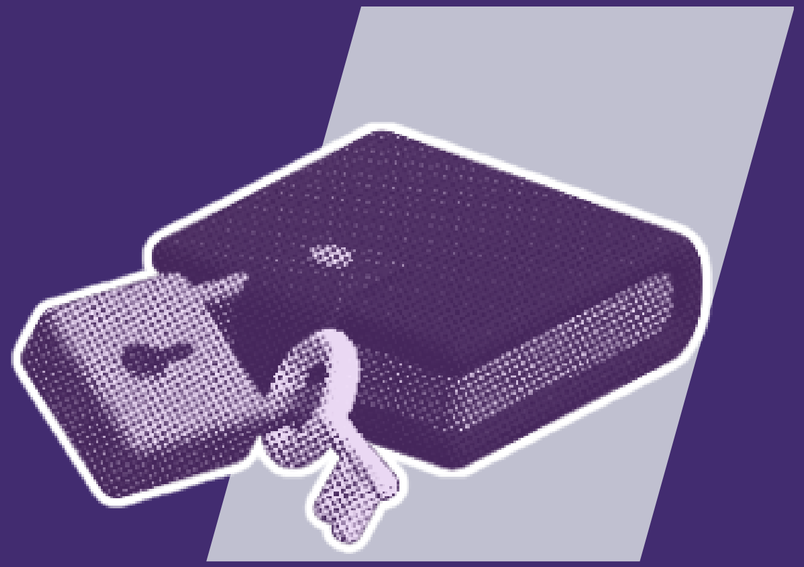
Introduction to Bitcoin Wallets

You have ownership of your bitcoin only if you control the private keys allowing you to sign transactions and transfer ownership of your bitcoin to someone else. This is the act of sending bitcoin.

Self - Custodial

vs

Custodial



Custodial

Benefits

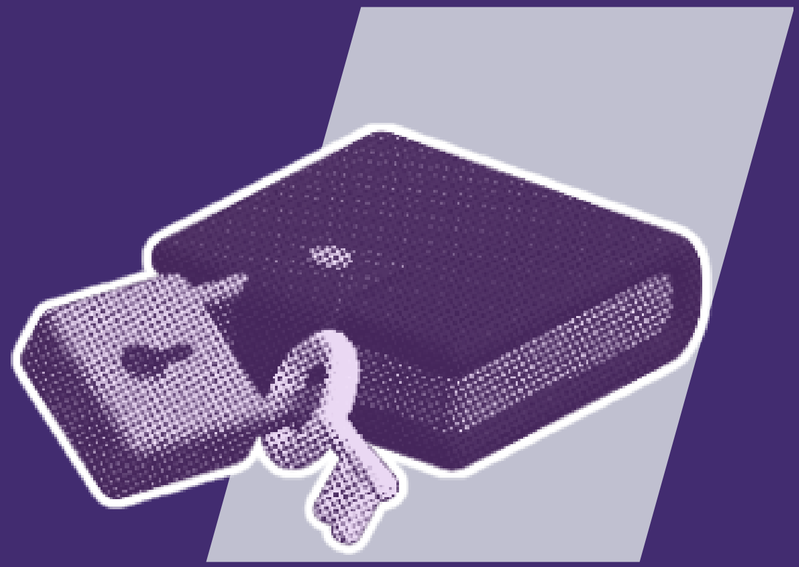
Easy recovery if access is lost, easier customer support.

Who controls my bitcoin?

The thirdparty provider

Risks

Custodial control means the custodian can freeze accounts.



Self - Custodial

Benefits

Complete control over funds and transactions, no approval process or account freeze, no corporate or government control.

Who controls my bitcoin?

The user

Risks

No recovery if recovery phrase is lost, less customer support.

“Not your keys, not your coins”

Different Types of Bitcoin Wallets

Online Wallet

Mobile Wallet

Desktop Wallet

Hardware Wallet

Online Wallet

Accessed through a web browser

Accessible from any device with an internet connection

Mobile Wallet

Installed on a mobile device

Easy to use

Desktop Wallet

Installed on a desktop computer

Convenient and can be accessed from anywhere

Hardware Wallet

A physical device that stores bitcoin offline

More secure than online wallets and can be used offline

What to consider when choosing a wallet

What to consider when choosing a wallet

Security: Make sure the wallet has strong security measures in place.

Privacy: Consider whether the wallet requires personal information.

Ease of use: Choose a wallet that is easy to use and navigate.

Compatibility: Make sure the wallet is compatible with your device.

Fees: Compare the fees charged by different wallets.

Reputation: Check the developers' reputation to ensure they are trustworthy.

Control: Some wallets give you more control over your private keys

Open Source vs Closed Source

Another important factor to keep in mind when choosing a Bitcoin wallet is knowing if the application or software is open-source. This is important because open-source projects let the community review the code and continue the project if the team stops working on it.

How to create a Bitcoin wallet

1

Search for the app in the App Store (iOS) or Google Play Store (Android)

2

You will be prompted to write down a list of 12 to 24 words and keep it in a safe place.

3

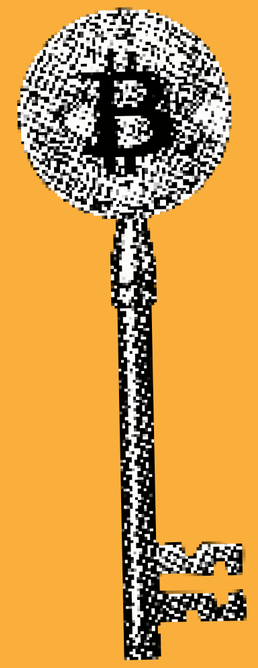
You must then confirm that you have saved your recovery phrase. To do this, you must enter, in the same order, the words of your seed phrase.

Once your recovery phrase is created and securely backed up, your wallet will be ready to send and receive bitcoin without intermediaries.

Key takeaways

1. Bitcoin can be acquired by earning it, mining it, buying it peer to peer, or purchasing it through exchanges.

2. Ownership of bitcoin depends on controlling the private keys associated with a wallet.



My
First
Bitcoin

Using Bitcoin in Daily Life

Module 07

“We’re building the Visa network for Bitcoin. But what I think is powerful is, unlike Visa, anybody can build on top of it.”

— Elizabeth Stark

The Lightning Network

The Lightning Network

Lightning Network is a payment system that allows users to send and receive bitcoin quickly and inexpensively. It works by setting up a shared wallet where both parties store some of their bitcoin. They can then make unlimited transactions with each other without needing to record each one on the main blockchain.

Lightning Network Transactions

On-Chain Transactions

Lightning Network Transactions

Speed

Instant. Payments settle in milliseconds, like a chat message.

Fees

Near Zero. Fees are just a few satoshis or even fractions of a satoshi.

Ideal Use

Daily Payments. Buying coffee, tips, or small transfers.

Security

High. Relies on smart channels; less robust for massive amounts

Lightning Network Transactions

On-Chain Transactions

On-Chain Transactions

Speed

Slow. Takes between 10 to 60 minutes (or more) to confirm.

Fees

Variable. Can be expensive depending on network congestion (Mempool).

Ideal Use

Store of Value. Large amounts, long-term savings (HODL).

Security

Maximum. Backed by the total computing power of all miners.

On-chain transactions happen directly on the Bitcoin blockchain. They are very secure but can be slower and more expensive.

Lightning transactions happen off-chain and allow payments to move much faster and at a much lower cost.

Sending and Receiving Lightning Transactions

Receiving payments

1

Click on "Receive" and a series of commands will appear.

2

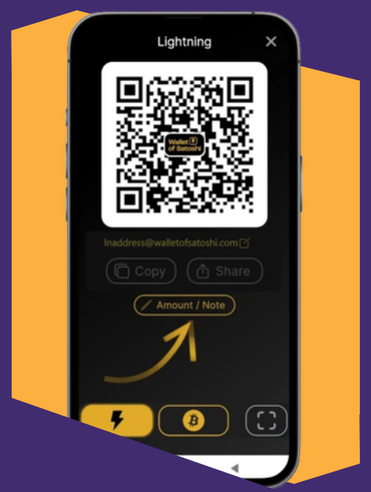
You can receive:

- Through LN-Address (a)
- Through LN, by adjusting the Invoice (b)
- On-chain (c)
- By scanning the QR code of an LNurl-p (d)

3

Click on "Receive" and choose the command with the Lightning Network symbol.

4



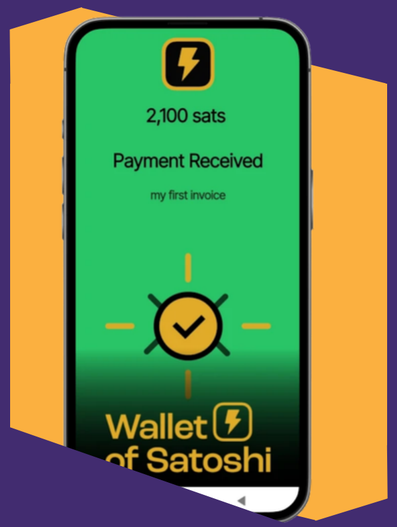
The Invoice creation menu will appear, where we click on "Add Amount" to type the exact amount.

Receiving payments

5

Using the keypad, we set the amount.

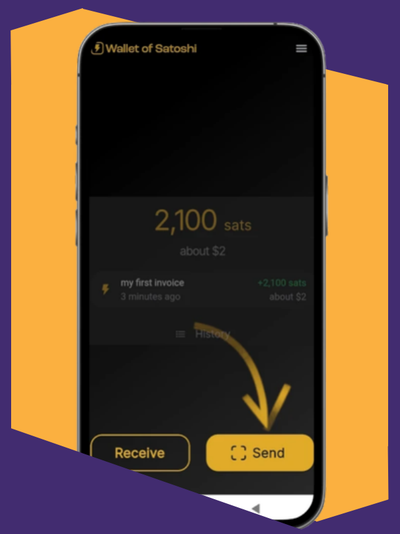
6



to then collect the Invoice. The received payment appears like this

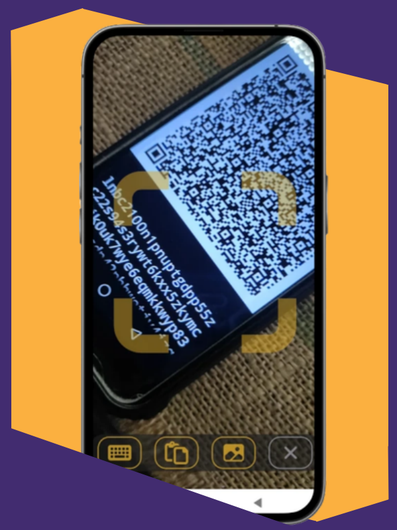
Sending payments

1



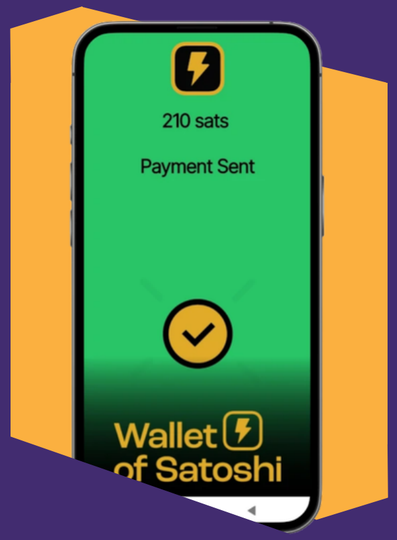
First, select the "Send" option

2



then scan the QR code and confirm the transaction

3

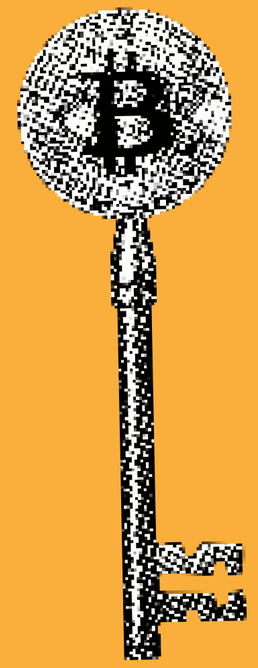


to finally send your payment.

Key takeaways

1. Lightning transactions happen off-chain, while on-chain transactions occur directly on the Bitcoin blockchain.

2. Bitcoin and Lightning enable everyday payments like coffee or groceries without traditional banking intermediaries.



My
First
Bitcoin

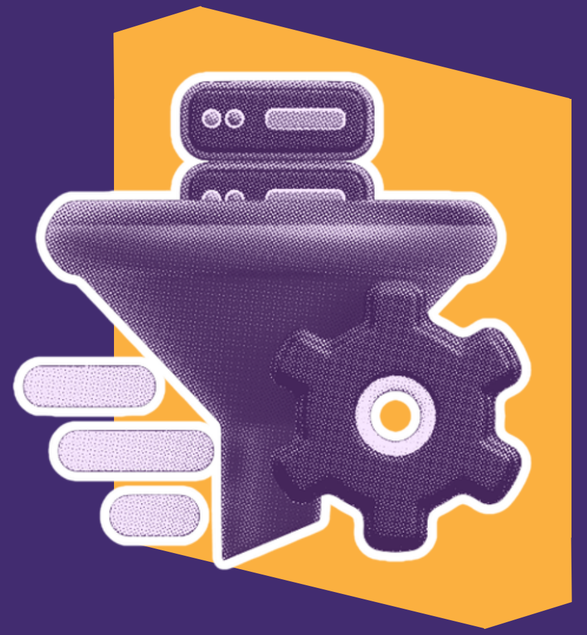
How Bitcoin works

Module 08

“Bitcoin is not "unregulated;" it is regulated by algorithm instead of being regulated by government bureaucracies. Uncorrupted.”

— Andreas M. Antonopoulos

How Bitcoin works



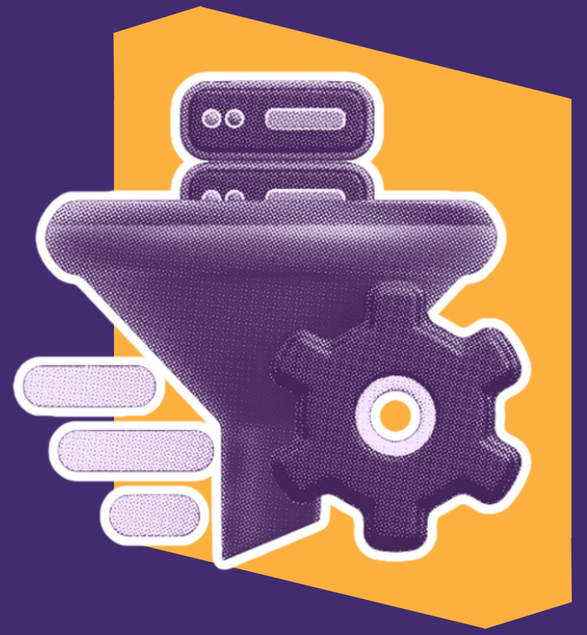
Mechanics of the Bitcoin Protocol

The Bitcoin network is a shared record of transactions stored across many computers called nodes. This record, known as the Bitcoin ledger, is pseudonymous. It does not include personal details like names or ages, only transaction data and Bitcoin addresses. The ledger tracks every transaction since the blockchain began.

Mechanics of the Bitcoin Protocol

- Proof-of-Work
- Hash Functions & Merkle Trees
- Peer-to-Peer Network Architecture
- Difficulty Adjustment
- Block Subsidy Halving
- Private and public key

How Bitcoin works



Security through Cryptography

Traditional Cryptography: Requires sharing the same secret key beforehand; if intercepted, security is completely compromised.

Bitcoin's Solution: Uses public-key cryptography, eliminating the need to exchange passwords in advance.

Public Key: Open and can be freely shared with anyone.
Private Key: A mathematical secret that must never be revealed.

Security through Cryptography

Private Key: Functions like your digital signature or digital password. It must be kept strictly secret and is used to authorize and sign your bitcoin transactions.

Public Key: Functions like your Bitcoin account number or address. It is safe to share with anyone so they can send you funds.

How a Bitcoin Transaction Works

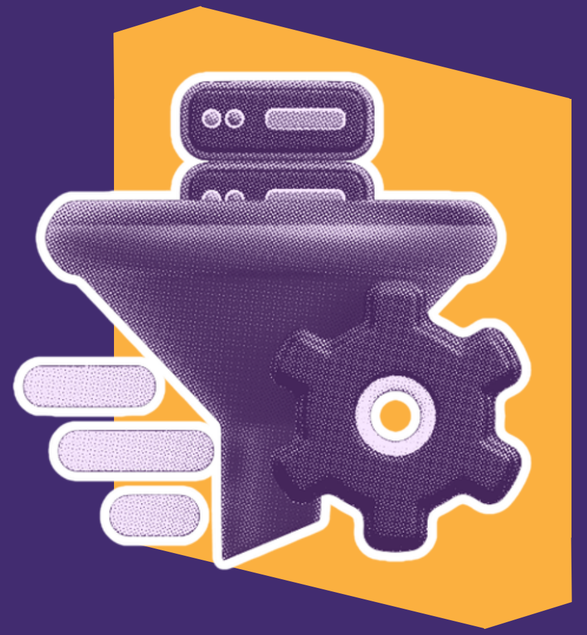
1. Create and Sign: The user generates the transfer and securely authorizes it using their private key.

2. Broadcast to Network: The transaction is broadcast globally to the interconnected network of P2P nodes.

3. Decentralized Validation: Each node independently verifies that the transaction complies with the rules (checking that funds exist, the signature is valid, and there is no double-spending).

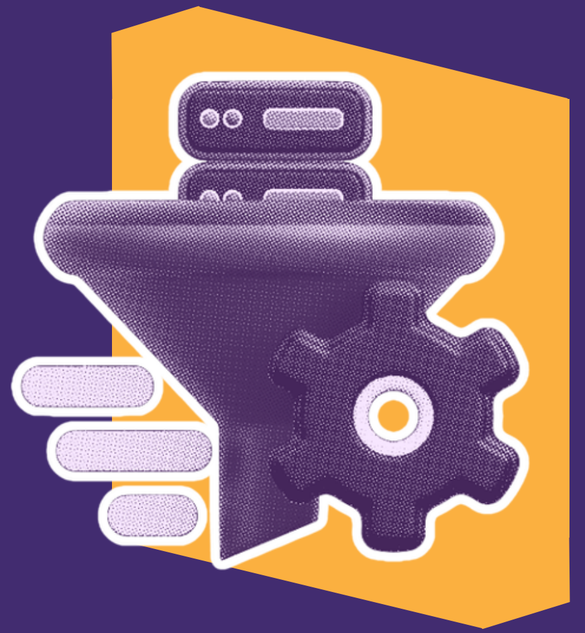
4. Store in the Mempool: The transaction is temporarily stored in each node's mempool (waiting room), ready to be included in the next block.

How Bitcoin works

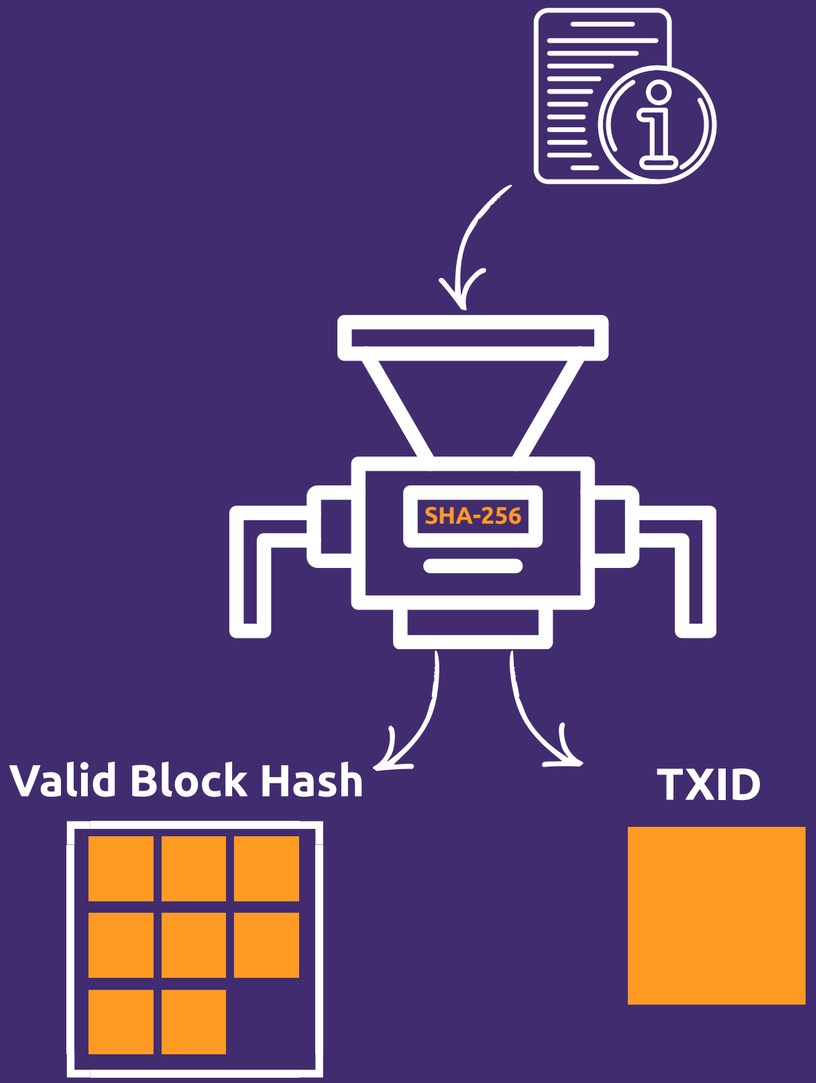


What is a function?

A hash function is like a secret code machine. It takes in a message and turns it into a code.



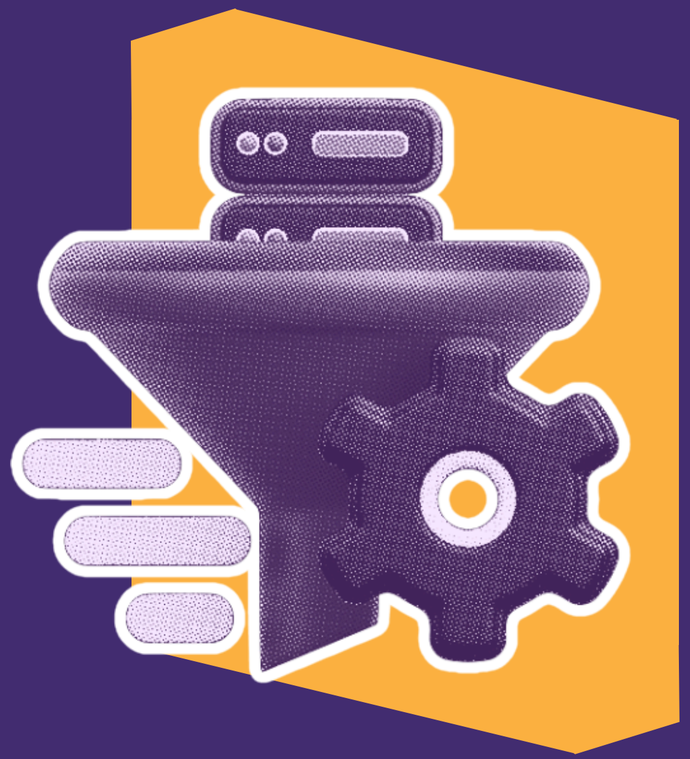
How Hashing Works in Bitcoin Transactions



- **Identify Transactions** (TXID): It turns transfer data into a unique fingerprint called a TXID, allowing any payment to be easily tracked and verified on the network.

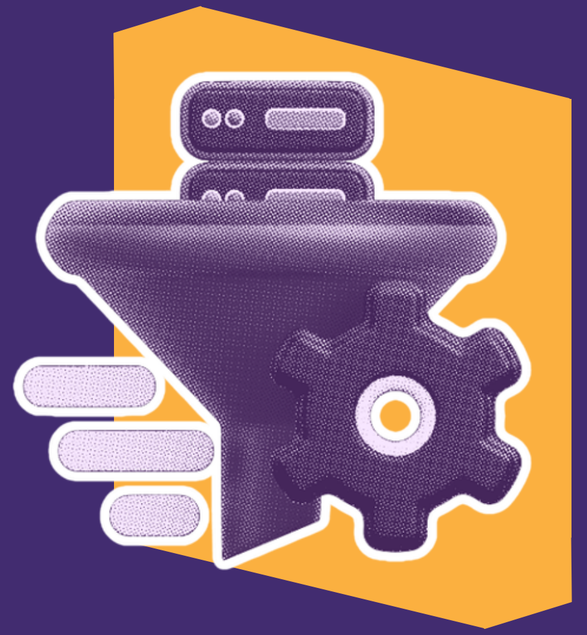
- **Secure and Link Blocks** (Valid Block Hash): Used in the mining process to generate a valid identifier for the entire block, locking it securely into the Blockchain history.

Get your own hash!



emn178.github.io/online-tools/sha256.html

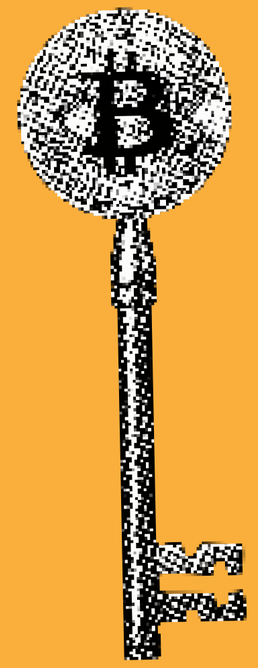
How Bitcoin works



Key takeaways

1. Bitcoin uses cryptography, including public and private keys, to securely authorize transactions.

2. Hash functions create unique digital fingerprints that protect data from tampering.



My
First
Bitcoin

How Does Bitcoin Mining Work?

Module 09

“The purpose of mining is not the creation of new bitcoin; that’s the incentive system. Mining is the mechanism by which Bitcoin’s security is decentralized.”

— Andreas M. Antonopoulos

Bitcoin Nodes and Miners

Miners

Nodes

Miners

Architects of security who solve cryptographic puzzles by using powerful computers and electricity. They check and confirm transactions, making sure everything is secure. Their computational work helps make the ledger (or blockchain) resistant to any bad by making the alteration of the blockchain expensive in energy terms.

Nodes

Gatekeepers of validation whose main job is to keep a copy of the Bitcoin ledger, and to make sure that all transactions are valid and that everyone follows the same rules. By distributing this job among many people worldwide, Bitcoin stays strong against potential fraud without relying on a centralized authority.

Miners

Miners compete to add the next block of transactions to the blockchain. To do this, they must find a special number that creates a valid block hash. You can imagine it like searching for the right key among billions of possibilities. The first miner to find the correct hash wins the race and earns the right to add their block to the blockchain

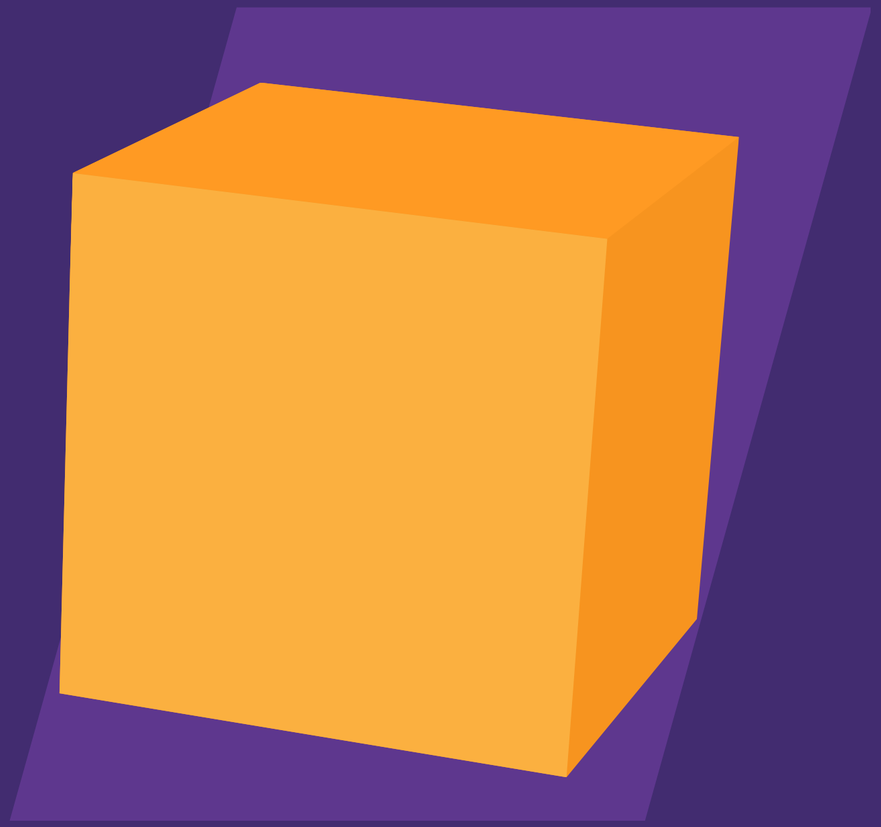
How Does Bitcoin Mining Work?

Valid Block Hash

Difficulty Adjustment

Reward

Valid Block Hash



It is the digital fingerprint of the block header. It includes the hash of the previous block, the Merkle root (all transactions), the nonce, and the timestamp. Its function is to chain the blocks together.

It must fulfill a mathematical condition: starting with a specific number of zeros (the difficulty target). Miners go through trillions of attempts by changing the nonce to achieve this.

Difficulty Adjustment

It is the automatic mechanism that keeps the issuance of Bitcoin constant, regardless of how many miners enter or leave the network.

The adjustment happens every 2,016 blocks or approximately every 2 weeks

Reward

Miners earn bitcoin in two ways:

Block rewards: New bitcoin are created and given to the miner who successfully adds a block to the blockchain.

Transaction fees: When people send bitcoin, they include a small fee. The miner who adds the block receives the fees from the transactions included in that block.

Bitcoin Nodes and Miners

1. Create and Sign: The user generates the transfer and securely authorizes it using their private key.

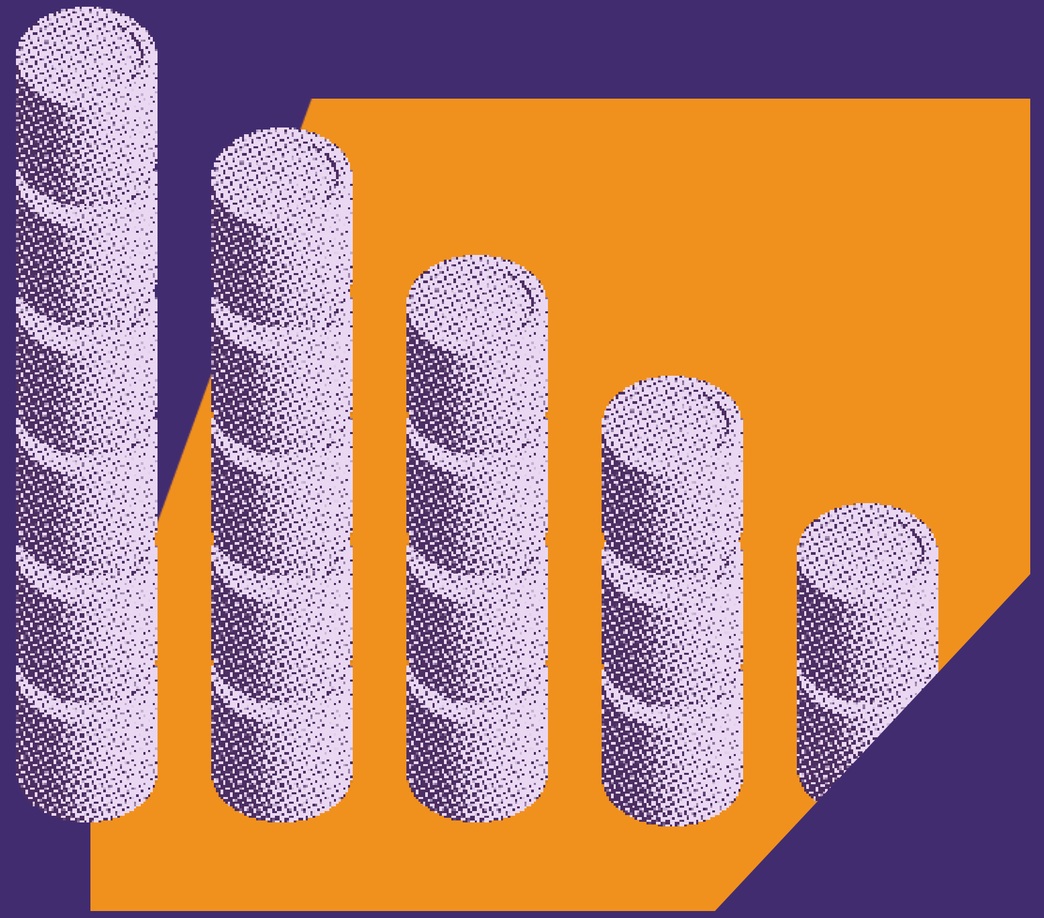
2. Broadcast to Network: The transaction is broadcast globally to the interconnected network of P2P nodes.

3. Decentralized Validation: Each node independently verifies that the transaction complies with the rules (checking that funds exist, the signature is valid, and there is no double-spending).

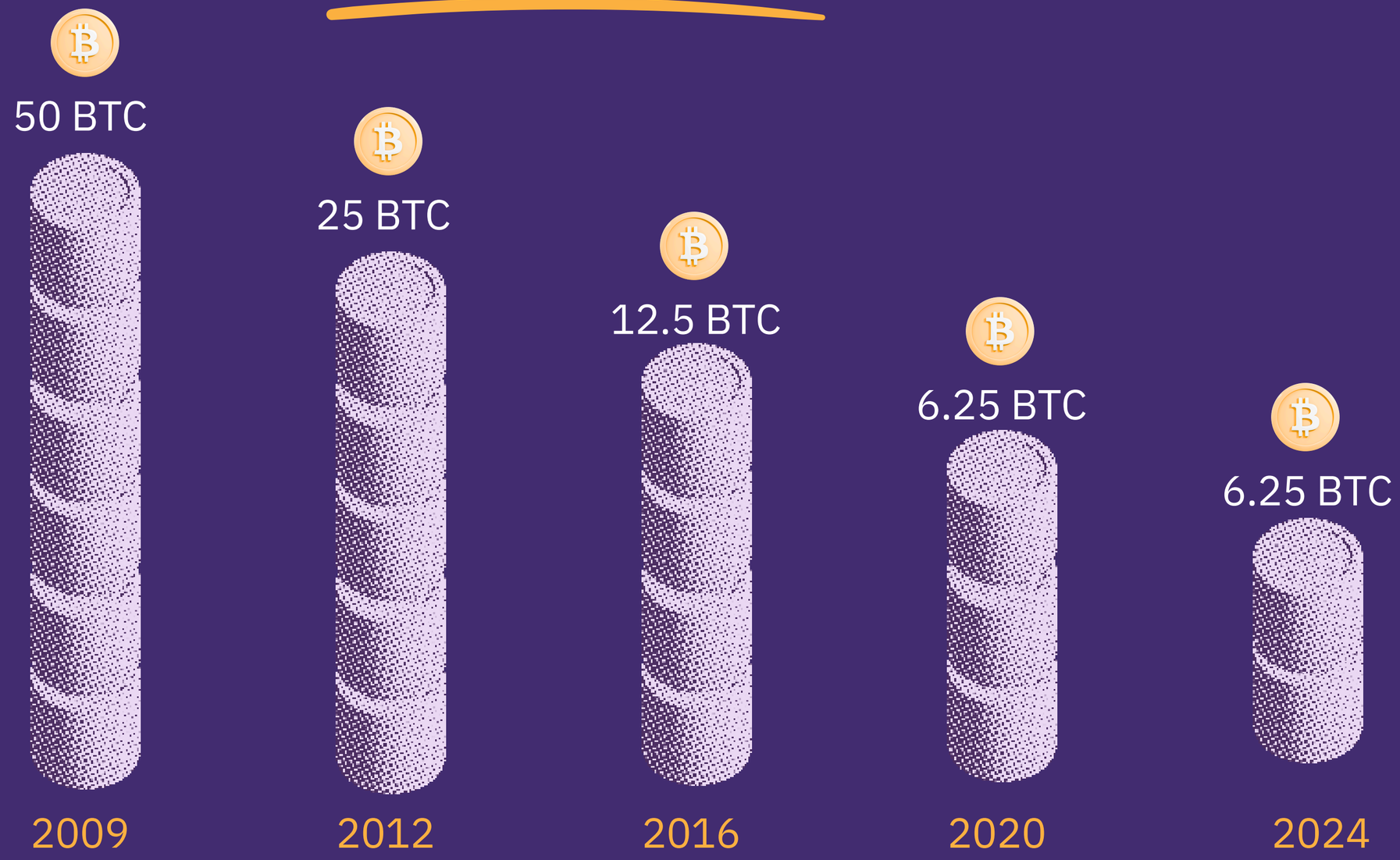
4. Store in the Mempool: The transaction is temporarily stored in each node's mempool (waiting room), ready to be included in the next block.

5. Mining Confirmation: Miners take transactions from the mempool, validate them again, and solve the cryptographic puzzle to add them permanently to a new block on the Blockchain.

Bitcoin Halvings



Bitcoin Halvings



What Is the Mempool?

The mempool, short for “memory pool,” is like a waiting room for Bitcoin transactions. When you send bitcoin, your transaction is first broadcast to the network and placed in the mempool.

You can think of it like waiting in line at a restaurant. Your name goes on a list, and you wait until a table is available. In the same way, your transaction waits in the mempool until a miner includes it in a block.

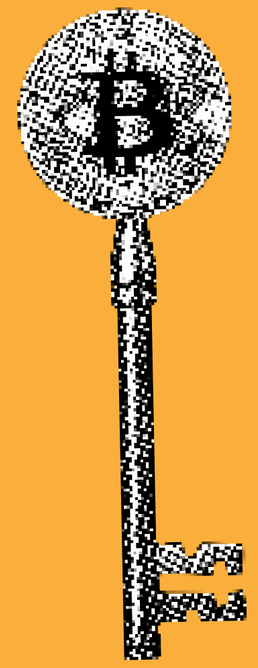
Activity: Exploring the Mempool

mempool.space

Key takeaways

1. Miners secure the network by performing Proof of Work to add new blocks.

2. Bitcoin transactions are signed, verified by nodes, mined into blocks, and permanently recorded on the blockchain.



My
First
Bitcoin

What Future Can Bitcoin Build?

Module 10

“Bitcoin is more than a currency; it's a revolution restoring power to the people, offering a taste of peace and freedom in a world hungry for empowerment.

— My First Bitcoin

What are Central Bank Digital Currencies (CBDCs)?

Central Bank Digital Currencies

Central Bank Digital Currencies, or CBDCs, are digital versions of regular fiat money. CBDCs follow the same rules as fiat money, where a central authority (like a government) can unilaterally expand the supply, reducing people's purchasing power in the process.

However, CBDCs also grant governments new and potent tools to control how that money is used by people around the world.

Central Bank Digital Currencies

cbdctracker.hrf.org

According to the Human Rights Foundation (HRF) research, 134 countries & currency unions, repres- enting 98% of global GDP, are exploring a CBDC. In May 2020 that number was only 35.

Currently, 66 countries are in the advanced phase of exploration—development, pilot, or launch. Every G20 country is exploring a CBDC, with 19 of them in the advanced stages of CBDC exploration.

Central Bank Digital Currencies

Bitcoin

Unlimited &
Centralized
Supply

Closed &
Permissioned

Total Control

Limited &
Decentralized
Supply

Open &
Permissionless

Absolute
Sovereignty

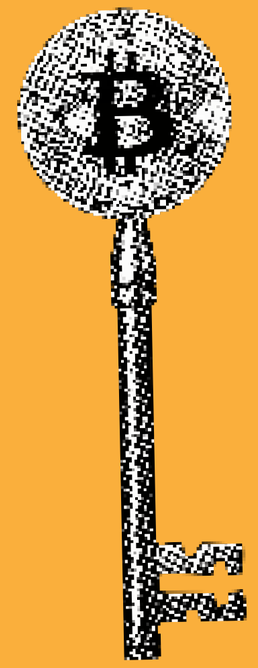
The Philosophy of Bitcoin

"Bitcoin belongs not to the powerful, but to the immutable rules we choose to protect together: it is the return of money to the harmony and collective sovereignty of humanity."

Key takeaways

1. Bitcoin is a decentralized monetary system with a fixed supply that no single authority can control

2. CBDCs are government-controlled digital currencies that allow centralized monitoring and control of transactions.



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